

Midterm Reviewer

Coverage: Modules 1–4 (L01–L04). Exam types: multiple choice, identification, solving (computation) and essay.

How to use it

1. **Study the weakest topic first.** The kit’s diagnostic ranks the topics; read in that order.
2. **Per topic:** intuition, then the notes, then the worked examples. Cover each solution and solve it yourself before reading it.
3. **Trust the refs.** Every fact carries a slide page. Slide errors are flagged; on the exam follow the slide’s intent but know the correct version.
4. **Use the cheat sheet pages** as the last-day summary, and the recall drill (the same sheet with its key terms blanked) to test yourself.
5. **Then simulate:** a mock exam on paper, timed, and mark it with the key.

Contents

Deck	Topic	Slides	In this reviewer
L01	Review of the Physical & Data Link Layers	18 slides	Included
L02	Signals, Bandwidth, Noise & Data-Rate Limits	39 slides	Included
L03a	Digital Transmission I: Line Coding, Block Coding & Scrambling	1–61	Included
L03b	Digital Transmission II: PCM, Delta Modulation & Transmission Modes	62–96	Included
L04	Digital-to-Analog: ASK, FSK, PSK & QAM	31 slides	Included

Pages: this cover, the cheat sheet, then one section per topic, each starting on a new page. Quick checks and practice are in the interactive kit, not in this reviewer.

NSCOM03 Midterm Cheat Sheet — Modules

L01–L04 · grey = slide page · bold = key term

L01 · Layers & data link

- OSI bottom-up: **Physical, Data link, Network, Transport, Session, Presentation, Application**
- Slide TCP/IP (4): Application = OSI 5–7 · Transport = 4 · **Internet = 3** · **Network access = 1–2**
- Slide 2 examples: switch, bridge, Ethernet, PPP → L2; hubs, repeaters, coax, fiber → L1; IP → L3; TCP/UDP → L4
- DLL: frames moved **error-free between two end nodes** over the physical layer
- DLL services: **framing**, frame sequencing, flow control, error detection (sometimes correction), **QoS**
- Physical layer: frames → electrical/optical/electromagnetic signals; cables, connectors, pins

L01 · Networks & signals

- LAN typically within **5 km**, copper/fiber, IEEE 802 · WAN: fiber, submarine cable, satellite & terrestrial microwave
- Dark fiber** = unused fiber · Starlink **550 km**, 775 satellites (Oct 2020)
- Wireless: unguided, cells; Bluetooth, 802.11, LTE
- IEEE 802: .2 LLC · .3 **Ethernet (CSMA/CD)** · .4 token bus · .5 token ring · .11 wireless · .13 none · .16 WLL
- MAC styles: **ALOHA style** vs **token style**
- Analog: vary amplitude/frequency/phase · Digital: e.g. 1 = +5 V, 0 = 0 V
- Analog transmission when **bandwidth is limited** → modulator / demodulator

L02 · Formulas

Frequency and period $f = 1/T$ · $T = 1/f$ f in Hz (cycles per second); T in seconds L02 p6; L02 p7

Wavelength $\lambda = c/f = c \times T$ c = propagation speed (3×10^8 m/s in free space) L02 p9

Bandwidth $B = f_{\text{high}} - f_{\text{low}}$ range of frequencies in a composite signal L02 p13

Bits per level bits per level = $\log_2 L$ L signal levels; bit rate = bits sent in 1 s L02 p14

Decibel $\text{dB} = 10 \log_{10}(P_2/P_1)$ negative = attenuated, positive = amplified; half power = -3 dB L02 p21; L02 p22

Signal-to-noise ratio $\text{SNR} = P_{\text{signal}}/P_{\text{noise}}$ · $\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$ average powers; SNR is a plain ratio L02 p25

Nyquist bit rate (noiseless) $\text{BitRate} = 2 \times B \times \log_2 L$ B = channel bandwidth (Hz); L = signal levels L02 p29

Shannon capacity (noisy) $C = B \times \log_2(1 + \text{SNR})$ SNR as a plain ratio, NOT dB; no levels in the formula L02 p31

Latency $\text{Latency} = T_{\text{prop}} + T_{\text{trans}} + T_{\text{queue}} + T_{\text{proc}}$ T_{prop} = distance / propagation speed; T_{trans} = message size / bandwidth L02 p35

L02 · Signals & bandwidth

- Analog = infinite levels · digital = limited levels; periodic vs nonperiodic
- $f = 1/T$ (Hz) · 4 cycles in 1 s → 4 Hz, $T = 1/4$ s · phase $0^\circ/90^\circ/180^\circ/270^\circ = 0^\circ/\text{peak}/0^\circ/\text{trough}$
- $\lambda = c/f$, $c = 3 \times 10^8$ m/s · red light 4×10^{14} Hz → **$0.75 \mu\text{m}$**
- Composite = sum of sines (Fourier); $B = f_{\text{high}} - f_{\text{low}}$ (5000 – 1000 = 4000 Hz)
- Digital signal = composite with **infinite bandwidth**; L levels → $\log_2 L$ bits per level
- Baseband → low-pass channel (first harmonic $B = N/2$) · broadband → modulation onto a **bandpass** channel

L02 · dB, SNR & noise

- $\text{dB} = 10 \log_{10}(P_2/P_1)$** ; loss negative, gain positive; half power = **-3 dB**
- dB of cascaded stages add; $P_{\text{out}} = P_{\text{in}} \times 10^{(\text{dB}/10)}$
- $\text{SNR} = P_s/P_n$** ; $\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$ · 10 mW / 1 μW → 10,000 → **40 dB**
- Noise: **thermal** (electrons), **induced** (motors), **crosstalk** (adjacent wire), **impulse** (spikes)
- Attenuation = energy loss · distortion = shape change

L02 · Rate limits & latency

- Nyquist (noiseless): **$N = 2B \log_2 L$** · 3000 Hz, $L = 2$ → **6000 bps**
- Shannon (noisy): **$C = B \log_2(1 + \text{SNR})$** , SNR linear · 3000 Hz, 3162 → **$\approx 34.86 \text{ kbps}$**
- Slide 33: 1 MHz, SNR 63 → **$C = 6 \text{ Mbps}$** ; use 4 Mbps → **$L = 4$**
- Latency** = $T_{\text{prop}} + T_{\text{trans}} + T_{\text{queue}} + T_{\text{proc}}$
- $T_{\text{prop}} = d/v$ · $T_{\text{trans}} = \text{size}/\text{bandwidth}$ · 12,000 km at 2.4×10^8 m/s → **50 ms**; 2.5 KB at 1 Gbps → **0.020 ms**
- Throughput < bandwidth · jitter = variation in delay

L03a · Formulas

Signal (baud) rate $S = c \times N \times 1/r$, N = data rate (bps); r = data elements per signal element; c = case factor (worst 1, best 0, average 1/2) L03 p8

Minimum bandwidth $B_{\text{min}} = S_{\text{avg}} \text{NRZ-L/NRZ-I/AMI N/2} \cdot \text{RZ, Manchester N} \cdot 2\text{B1Q N/4} \cdot \text{MLT-3 N/3} \cdot 8\text{B6T 3N/4} \cdot 4\text{D-PAM5 N/8}$ L03 p25; L03 p46

mBnL code condition $2^m \leq L^n$ m data bits → n signal elements with L levels ($B = 2$, $T = 3$, $Q = 4$) L03 p36; L03 p37

4B/5B rate $N_{\text{line}} = N \times 5/4$ every 4 data bits become 5 code bits L03 p54

L03a · Rates & properties

- $S = c \times N \times 1/r$** (baud); c = 1/2 average; $B_{\text{min}} = S$
- r: NRZ 1 · RZ/Manchester 1/2 · **2B1Q 2** · 8B6T 4/3
- Avg B: NRZ-L/I, AMI **N/2** · RZ, biphasic **N** · 2B1Q **N/4** · MLT-3 **N/3** · 8B6T **3N/4** · 4D-PAM5 **N/8**
- Good code: no **baseline wandering**, no **DC**, self-sync, error detection, noise immunity, low complexity
- Ex 1: 100 kbps, $r = 1$ → **50 kbaud** · Ex 2: 0.1% fast → 1001 bps / 1,001,000 bps · Ex 3: NRZ-I 1 Mbps → **500 kbaud, 500 kHz**

L03a · Scheme rules (slide figures)

- Unipolar NRZ 1 = +V, 0 = 0 · **NRZ-L 0 = +V, 1 = -V** · **NRZ-I: 1 inverts** (start +)
- RZ: 1 = +V→0, 0 = -V→0 · **Manchester 0 = high→low, 1 = low→high**
- Diff. Manchester: 0 = transition at start**, always mid-bit transition
- AMI 0 = 0 V, 1s alternate** (first +) · pseudoternary = reverse (low priority)
- 2B1Q (prev +): 00 +1, 01 +3, 10 -1, 11 -3 (prev -): signs flip)
- MLT-3: 0 stays; 1 → 0 from ±V, or → opposite of last nonzero from 0
- mBnL needs **$2^m \leq L^n$** ; B = 2, T = 3, Q = 4 levels

L03a · Block coding & scrambling

- Block coding mB/nB: division → substitution → combination; adds redundancy
- 4B/5B + NRZ-I**; rate $\times 5/4$; 16 spare codes (control: I 11111, Q 00000, H 00100, J 11000, K 10001, T 01101)
- Ex 4: 1 Mbps → **1.25 Mbps** → NRZ-I **625 kHz** / Manchester 1.25 MHz
- 8B/10B = 5B/6B + 3B/4B + disparity controller
- B8ZS: 8 zeros** → **000VBOVB** (V same polarity as previous pulse)
- HDB3: 4 zeros** → **B00V** if even, **000V** if odd (pulses since last substitution)

L03b · Formulas

Sampling rate $f_s = 1/T_s$ T_s = sampling interval (s); f_s in samples/s L03 p64

Nyquist sampling theorem $f_s \geq 2 \times f_{\text{max}}$ f_{max} = highest frequency in the signal — for low-pass and bandpass signals alike L03 p66; L03 p67

Zone height (quantization step) $\Delta = (\text{max} - \text{min}) / L$ L = number of zones (levels) L03 p72

Bits per sample $n_b = \log_2 L$ L is a power of 2 in the slides; otherwise round up (textbook) L03 p75

PCM bit rate $N = n_b \times f_s$ bits/sample $\times$ samples/s L03 p79

Minimum PCM bandwidth $B_{\text{min}} = n_b \times f_{\text{max}}$ NRZ-type line code ($c = 1/2$, $r = 1$): $B_{\text{min}} = N/2$ Slide example 10: $8 \times 4 \text{ kHz} = 32 \text{ kHz}$. L03 p83

Quantization error bound $|\text{error}| \leq \Delta / 2$ error = quantized (midpoint) – actual L03 p76; L03 p78

L03b · PCM

- PCM = **sample** → **quantize** → **encode**; filter first
- $f_s = 1/T_s$; Nyquist $f_s \geq 2 \times f_{\text{max}}$ (low-pass and bandpass); voice 4 kHz → **8000 sps**
- Bandpass with only B known → rate **cannot be determined**
- Sampling: ideal (impulse) · natural (follows curve) · **flat-top** (sample and hold) → PAM
- $\Delta = (\text{max} - \text{min})/L$; zones 0 ... L–1 from the bottom; sample → **midpoint**; $n_b = \log_2 L$
- 20 ... +20 V, $L = 8$ → $\Delta = 5 \text{ V}$; 000 = -20 to -15 V ... 111 = +15 to +20 V
- $|\text{error}| \leq \Delta/2$; low amplitudes suffer → non-linear quantization, **companding**

L03b · Rates, DM, modes

- $N = n_b \times f_s$ — voice, 8 bits $\rightarrow$ **64 kbps**
- $B_{\min} = n_b \times f_{\max}$ (NRZ-type) — $8 \times 4 \text{ kHz} = \mathbf{32 \text{ kHz}}$
- Decoder: **hold circuit** + low-pass filter (cutoff = $f_{\max}$)
- DM: 1 bit per sample, 1 = up δ , 0 = down δ ; DPCM = quantized difference
- Parallel: n bits per tick on n wires · Serial: 1 bit per tick
- Async: **start 0** + byte + **stop 1**, gaps OK · Sync: no start/stop, frames · Isochronous: equal gaps

L04 · Formulas

Baud (signal) rate $S = N \times 1/r$ S in baud; N = bit rate (bps); r = data bits per signal element Analog transmission of digital data: $S \leq N$. L04 p5

Bits per element $\leftrightarrow$ levels $r = \log_2 L \Leftrightarrow L = 2^r$ L = number of distinct signal elements (levels, phases, frequencies) L04 p7

ASK / PSK bandwidth $B = (1 + d) \times S$ $0 \leq d \leq 1$ (modulation and filtering); d = 0 gives the minimum, B = S L04 p9; L04 p21

Binary FSK bandwidth $B = (1 + d) \times S + 2\Delta f$ $\Delta f = f_1 - f_2$; bit 1 $\rightarrow f_c + \Delta f$, bit 0 $\rightarrow f_c - \Delta f$ L04 p14; L04 p15

Multilevel FSK bandwidth $B = (1 + d) \times S + (L - 1) \times 2\Delta f$ with d = 0 and the minimum spacing $2\Delta f = S$: $B = L \times S$ Slide 18 prints “(L - 1)/2 Δf ” — it is a product. L04 p18; L04 p19

Carrier in an available band $f_c = (f_{\text{low}} + f_{\text{high}}) / 2$ full duplex: split the band in two halves, one carrier in the middle of each L04 p12; L04 p13

L04 · Digital $\rightarrow$ analog

- **ASK** amplitude · **FSK** frequency · **PSK** phase · **QAM** amplitude + phase
- $S = N/r$ (baud); $r = \log_2 L$, $L = 2^r$; $S \leq N$
- ASK/PSK: $B = (1 + d)S$, $0 \leq d \leq 1$; BASK = OOK, r = 1
- FSK: $B = (1 + d)S + 2\Delta f$; bit 1 $\rightarrow f_c + \Delta f$, bit 0 $\rightarrow f_c - \Delta f$
- MFSK: $B = (1 + d)S + (L - 1) \cdot 2\Delta f = L \cdot S$ when d = 0, $2\Delta f = S$
- Carrier at band middle; full duplex $\rightarrow$ halve the band (200–300 kHz $\rightarrow$ 225 and 275 kHz)
- BASK = **unipolar** NRZ $\times$ carrier; BPSK = **polar** NRZ $\times$ carrier; FSK via VCO
- PSK more robust than ASK (noise hits amplitude)

L04 · Constellations & examples

- x = I (in-phase), y = Q (quadrature); length = amplitude, angle = phase
- OOK: 0 at origin, 1 on +I · BPSK: 0 at -I, 1 at +I
- QPSK (1st bit $\rightarrow$ I): **11 $\rightarrow 45^\circ$, 01 $\rightarrow 135^\circ$, 00 $\rightarrow -135^\circ$, 10 $\rightarrow -45^\circ$**
- 16-QAM (slide 31): first 2 bits = quadrant (00 I, 10 II, 11 III, 01 IV); 1100 $\rightarrow 225^\circ$
- r = 4, S = 1000 $\rightarrow$ **4000 bps** · N = 8000, S = 1000 $\rightarrow$ **r = 8, L = 256**
- ASK 200–300 kHz, d = 1 $\rightarrow$ **250 kHz, 50 kbps** · FSK, $2\Delta f = 50 \text{ kHz} \rightarrow$ **25 kbps**
- QPSK 12 Mbps, d = 0 $\rightarrow$ **6 MHz** · MFSK 3 bits, 3 Mbps $\rightarrow$ **L = 8, 1 Mbaud, 8 MHz**

Symbol clashes

- **S = signal (baud) rate** in baud: $S = c \cdot N/r$ (L03), $S = N/r$ (L04). **C = Shannon capacity** in bps. Not the same thing.
- **c = case factor** (0 to 1, average $1/2$) in $S = c \cdot N/r$. In L02, **c = speed of light**, $3 \times 10^8 \text{ m/s}$.
- **N = bit (data) rate** in bps. $N_{\max} = 2 \cdot B \cdot \log_2 L$ is Nyquist's limit for a noiseless channel.

Mental-math tips (no calculator)

- $\log_2$ of a power of 2 = **count the doublings** ($\log_2 64 = 6$); $\log_2 1000 \approx \mathbf{10}$
- dB $\rightarrow$ ratio: **$10^{(\text{dB}/10)}$** ; ratio $\rightarrow$ dB: $10 \cdot \log_{10}(\text{ratio})$. Add dB to multiply ratios.
- **3 dB $\approx \times 2$** · **10 dB = $\times 10$** · 20 dB = $\times 100$ · -3 dB $\approx \times 1/2$
- **$2^{10} \approx 10^3$** (1024 vs 1000)
- Powers of 2: $2^1=2$ · $2^2=4$ · $2^3=8$ · $2^4=16$ · $2^5=32$ · $2^6=64$ · $2^7=128$ · $2^8=256$ · $2^9=512$ · $2^{10}=1024$
- $\text{SNR} = 2^k - 1$ gives $\log_2(1 + \text{SNR}) = k$: SNR 3, 7, 15, 31, 63, 255 $\rightarrow$ 2, 3, 4, 5, 6, 8

Unit traps

- **K = 1000** for data rates and sizes (1 kbps = 1000 bps; 1 KB = 1000 B on the slides), not 1024.
- Bytes $\times 8$ = bits (2.5 KB = 20,000 bits) before dividing by a bit rate.
- **$\mu\text{s} = 10^{-6} \text{ s}$, $\text{ms} = 10^{-3} \text{ s}$** : 0.020 ms = 20 μs .
- Convert everything to base units (Hz, bps, s, m) before substituting.

Review of the Physical & Data Link Layers

Where this course lives in the OSI and TCP/IP stacks, what the data link and physical layers do, the kinds of networks, and why digital data is sometimes sent as an analog signal.

★ OSI ↔ TCP/IP mapping

★ Data link layer services

★ LAN ≤ 5 km · WAN · wireless

★ IEEE 802 series

★ Why analog transmission

AT A GLANCE

- OSI has **7 layers** (bottom-up: Physical, Data link, Network, Transport, Session, Presentation, Application). The slides' TCP/IP model has **4**: Network access (OSI 1–2), Internet (3), Transport (4), Application (5–7). This course lives in the **bottom two**.
- The **data link layer** formats data into **frames** and moves them error-free between two end nodes over the physical layer. Its services: framing, frame sequencing, flow control, error detection (sometimes correction), and QoS.
- **LAN**: typically within **5 km**, copper or fiber, IEEE 802 standards. **WAN**: long distance — fiber (dark fiber = unused), submarine cable, satellite and terrestrial microwave. **Wireless**: unguided media, short or long range, organized in cells.
- **IEEE** took over LAN standards (physical + data link layers) using OSI as the framework: 802.3 Ethernet (CSMA/CD), 802.5 token ring, 802.11 wireless LAN, 802.16 wireless local loop.
- **Media access control (MAC)** coordinates nodes on a shared medium — **ALOHA style** or **token style**.
- The **physical layer** turns frames into electrical, optical or electromagnetic signals. **Analog transmission** is needed when bandwidth is limited: a **modulator** turns digital into analog and a **demodulator** turns it back.

WHAT THEY'LL ASK

- **MCQ**: which layer does what; which protocol or device sits at which layer; the TCP/IP ↔ OSI mapping; LAN distance; IEEE 802 numbers.
- **Identification**: dark fiber, framing, frame sequencing, QoS, IEEE, physical layer, modulator/demodulator, cell.
- **Essay**: explain the purpose and services of the data link layer; why analog transmission is used; compare LAN, WAN and wireless networks.
- **MCQ**: Starlink facts (550 km; 775 satellites by October 2020); ALOHA style vs token style MAC.

💡 INTUITION FIRST

Layers are like a postal system

OSI model (slide 2)

7	Application	End-user layer · HTTP, FTP, DNS
6	Presentation	Syntax layer · SSL, JPEG, MPEG
5	Session	Synch & send to port · sockets
4	Transport	End-to-end connections · TCP, UDP
3	Network	Packets · IP, ICMP
2	Data link	Frames · Ethernet, PPP, switch, bridge
1	Physical	Physical structure · coax, fiber, hubs

TCP/IP model (slide 3)

Application
Transport
Internet
Network access

The two bottom layers (highlighted) are the ones this deck reviews: the data link and physical layers.

A letter goes into an envelope, the envelope into a mailbag, the mailbag onto a truck. Each layer wraps the one above and only talks to its own counterpart at the other end. The **data link layer** is the hand-off between two *neighbouring* post offices (one hop, in frames); the **physical layer** is the road itself — the signals on the wire.

Predict: A switch forwards frames between two computers on the same LAN. Which OSI layer is it working at?

Answer: Data link

Switches and bridges handle **frames**, so they work at the data link layer — slide 2 lists them there. Hubs and repeaters only repeat signals (physical layer).

Predict: A medium only passes a limited band of frequencies, so a square digital signal comes out badly distorted. What do the slides prescribe?

Answer: Modulate it onto an analog signal (modulator/demodulator)

Slide 17: analog transmission is needed when bandwidth is limited — the digital signal is modulated into an analog signal, transmitted, and demodulated at the other end.

The OSI and TCP/IP models L01 pp2–3

The slide's OSI table (a generic web graphic) gives each layer a one-line role and examples: L01 p2

#	OSI layer	Role (slide)	Examples (slide)
7	Application	End-user layer	HTTP, FTP, IRC, SSH, DNS
6	Presentation	Syntax layer	SSL, SSH, IMAP, FTP, MPEG, JPEG
5	Session	Synch & send to port	APIs, sockets, WinSock
4	Transport	End-to-end connections	TCP, UDP
3	Network	Packets	IP, ICMP, IPSec, IGMP
2	Data link	Frames	Ethernet, PPP, switch, bridge

#	OSI layer	Role (slide)	Examples (slide)
1	Physical	Physical structure	Coax, fiber, wireless, hubs, repeaters

The slides compare it with a **4-layer TCP/IP (Internet) model**: **Application** = OSI 7, 6 and 5; **Transport** = OSI 4; **Internet** = OSI 3; **Network access** = OSI 2 and 1. L01 p3

⚠ SLIDE SAYS / CORRECT

Slide 2 FTP and SSH appear at two layers; IMAP and SSL sit at Presentation.

Correct FTP, SSH and IMAP are application-layer protocols. Know the slide's table for the exam, but don't be surprised if a question treats them as application protocols.

Slide 3 TCP/IP has 4 layers (Network access = OSI 1–2).

Also The textbook (Forouzan) uses a 5-layer TCP/IP model — physical, data link, network, transport, application. Same functions, different grouping.

The data link layer L01 pp4–5

Purpose: to regulate and format the transmission of data from software on a node to the network cabling. It creates the network environment for the “wire” and dictates data formats, timing, bit sequencing and other activities for each type of network, so that **data frames are transmitted error-free between two end nodes over the physical layer**. L01 p4

Services it gives the upper layers L01 p5 :

- **Provisioning links** between network entities — generally adjacent nodes within a subnetwork.
- **Framing** — partitioning data into frames with recognized frame boundaries and exchanging them over the link.
- **Frame sequencing** — keeping frames in the correct order as they are exchanged.
- **Flow control** — establishing and maintaining an acceptable flow as frames cross the link.
- **Error detection** — detecting physical-layer errors, with notification when errors are detected but not corrected; it **can sometimes correct** errors too.
- **Quality of service (QoS)** — selecting QoS parameters: sufficient bandwidth available, and transmission delays that are predictable and guaranteed.

Types of networks L01 pp6–12

The data link layer serves three kinds of networks: L01 p6

Type	What the slides say
LAN — local area network	Typically within 5 km ; computers are normally adjacent; interconnected with copper or fiber ; typically uses the IEEE 802 series. L01 p7
WAN — wide area network	Geographically long-distance connections: literally a long cable, normally fiber optic ; microwave via satellites or terrestrial links. L01 p10
Wireless network	Unguided media; short or long distance; radio signals (also microwave). L01 p12

IEEE and the LAN

In the early days there were no standards: organizations used proprietary networks and were locked into a vendor or technology. The **IEEE** took responsibility for setting LAN standards, primarily for the **physical and data link layers**, using the OSI reference model as a framework. L01 p8

Standard	Defines (slide 9)
802.1	Architectural overview of LANs
802.2	Logical Link Control (LLC)
802.3	CSMA/CD — Ethernet (802.3u Fast Ethernet 100 Mbps; 802.3z/802.3ab Gigabit; 802.3ae 10-Gigabit)
802.4	Token bus
802.5	Token ring (logical ring, token passing)
802.6	Metropolitan area networks (MANs)
802.7	Broadband LANs (video, data, voice)
802.9	Integrated Services LANs (ISLANs)
802.10	LAN/MAN security
802.11	Wireless media access control and physical layer
802.12	Demand priority (100VG-AnyLAN)
802.13	Nothing — skipped over superstition about “13”
802.14	Cable-TV broadband
802.16	Wireless local loop (WLL)

WAN media

- **Fiber optic** is the norm. **Dark fiber** = unused fiber-optic cable. Examples: **submarine cable**, fiber laid beside roads or commuter-train railways. L01 p10
- **Microwave: satellites** orbiting the earth (uplink and downlink between ground stations) and **terrestrial microwave links**. L01 p10
- **SpaceX Starlink**: satellites deployed globally at **550 km** above the earth to bring high-speed broadband where access is unreliable, expensive or unavailable; **775 satellites** launched as of October 2020. L01 p11

Wireless networks

Use radio signals (also microwave); can be a personal, local or wide area network; use a “**cell**” to interconnect mobile devices. Examples: **Bluetooth**, the **IEEE 802.11** series, the **LTE** mobile phone network. The slide’s cellular diagram shows a call being set up through the switch (MTSO): it verifies both phones as valid subscribers, finds voice channels, locates phone B’s strongest cell, connects the voice path, and monitors the call for hand-off. L01 p12

Media access control (MAC) L01 p13

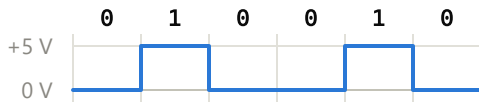
The MAC layer establishes **coordination between nodes** so that a node can gain access to the shared medium and transmit. There are different techniques; the slides name two families, **ALOHA style** and **token style**. L01 p13 **Beyond slides** In the textbook, ALOHA style means contention (stations transmit and deal with collisions) and token style means a station may send only while it holds the token.

The physical layer and signal transmission L01 pp14–17

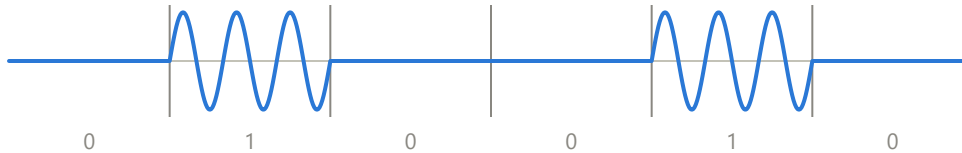
The **physical layer** translates frames from the data link layer into **electrical, optical or electromagnetic signals** representing 0s and 1s. Its specification covers the type of cable and connectors, the electrical signal on each pin, and how bit values become physical signals. L01 p14

- **Analog communication:** the signal varies continuously in strength or quantity (e.g. voltage); data is represented by varying the **voltage** (amplitude), the **frequency** or the **phase** of a wave. L01 p15
- **Digital communication:** the signal is coded in binary — e.g. a 1 as **+5 V** and a 0 as **0 V**. L01 p15

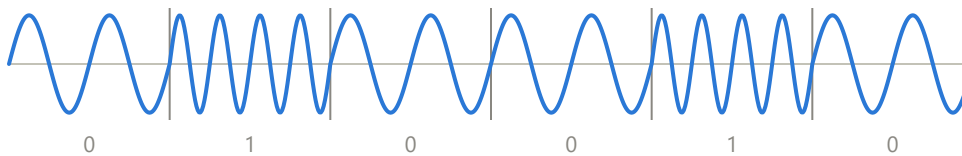
Digital transmission: 1 = +V, 0 = 0 V



Analog transmission: amplitude changes (AM) ...



... or frequency changes (FM)



Slide 16: analog transmission varies a carrier (here amplitude or frequency); digital transmission sends voltage levels.

Why use analog transmission?

Analog transmission is needed **when bandwidth is limited**: a medium with limited bandwidth does not let a digital signal pass through, so digital signals are **modulated** into analog signals and transmitted over the medium, then **demodulated** at the far end. For two-way traffic, each direction has its own modulator and demodulator. L01 p17

CONNECTS TO

L02 explains *why* limited bandwidth distorts a digital signal (low-pass vs bandpass channels), and L04 covers the modulation schemes (ASK, FSK, PSK, QAM).

WORKED EXAMPLE

Placing devices and protocols on the stack L01 p2; L01 p3

Task: say which OSI layer, and which slide TCP/IP layer, each item belongs to: *switch, repeater, TCP, IP, HTTP, Ethernet*.

1. **Repeater** — only regenerates signals → OSI 1 Physical → TCP/IP Network access.
2. **Switch** and **Ethernet** — work with frames → OSI 2 Data link → Network access.
3. **IP** — packets → OSI 3 Network → Internet.
4. **TCP** — end-to-end connections → OSI 4 Transport → Transport.
5. **HTTP** — end-user application → OSI 7 Application → Application.

⚠ COMMON TRAPS

- **Mixing up the order of the layers** → Bottom-up: Physical, Data link, Network, Transport, Session, Presentation, Application ("Please Do Not Throw Sausage Pizza Away"). L01 p2
- **Mapping TCP/IP's Internet layer to OSI 2** → Internet = OSI 3 (Network). Network access = OSI 1–2. L01 p3
- **Saying a LAN spans tens of kilometres** → The slides: typically within 5 km. L01 p7
- **Thinking "dark fiber" is a special kind of fiber** → It is simply unused (unlit) fiber-optic cable. L01 p10
- **Saying analog is used because it is faster** → It is used when bandwidth is limited and a digital signal cannot pass through the medium. L01 p17
- **Putting routing in the data link layer** → The data link layer works between adjacent nodes (frames); routing packets between networks is the network layer's job. L01 p5

💡 MNEMONICS & MEMORY HOOKS

- Bottom-up OSI: "Please **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way". Top-down: "**A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing".
- TCP/IP on the slides: 4 layers cover the 7 OSI layers as 3 + 1 + 1 + 2 — Application 3, Transport 1, Internet 1, Network access 2.
- DLL services — "**F**ramed **S**equences **F**low **E**rror-free with **Q**uality": framing, sequencing, flow control, error detection, QoS.
- IEEE 802 favourites: **.3** Ethernet, **.5** token ring, **.11** Wi-Fi, **.13** nothing (unlucky).
- Starlink: "**550** up, **775** launched by October 2020".

Blank-page recall

Close the notes. Answer each prompt from memory on paper, then check against the sections above.

1. Draw the OSI stack and the slides' 4-layer TCP/IP model side by side, with the mapping.
2. State the purpose of the data link layer and list its six services.
3. Compare LAN, WAN and wireless networks: distance, media, examples.
4. Why did IEEE take over LAN standards, and which layers do they cover? Name four 802 standards.
5. Name the two styles of media access control.
6. Explain why digital data is sometimes sent as an analog signal.

⚠ SLIDE ERRORS & CONTRADICTIONS

On the exam, the answer that follows the slide's intent is the safe one — but know the correct version.

Slide	Slide says	Correct / intended
L01 p2	FTP and SSH appear under both Application and Presentation; IMAP and SSL under Presentation	FTP, SSH and IMAP are application-layer protocols (the table is a generic web graphic). Learn the slide's table, but expect them to be called application protocols elsewhere.
L01 p3	A 4-layer TCP/IP model	The textbook (Forouzan) uses 5 layers: physical, data link, network, transport, application.
L01 p9	The 802 list skips 802.8 and 802.15 and calls 802.3c "10 Mbps Ethernet"	802.15 is the wireless PAN group (Bluetooth) and 802.8 was the fiber-optic advisory group; 10BASE-T is 802.3i.
L01 p10	The "terrestrial microwave" picture shows ground-wave and sky-wave (ionosphere) paths	Terrestrial microwave is line-of-sight between antennas on towers.
L01 p16	"Multipelxed Sum"	Typo for "multiplexed".

Glossary

OSI model	Seven-layer reference model: physical, data link, network, transport, session, presentation, application. L01 p2
TCP/IP model	On the slides, 4 layers: application (OSI 5–7), transport (4), internet (3), network access (1–2). L01 p3
Data link layer	Formats data into frames and moves them error-free between two end nodes over the physical layer. L01 p4
Framing	Partitioning data into frames with recognized boundaries and exchanging them over the link. L01 p5
Frame sequencing	Keeping frames in the correct order as they are exchanged. L01 p5
Flow control	Keeping the flow of frames across a link at an acceptable level. L01 p5
Quality of service (QoS)	Transmission parameters such as sufficient bandwidth and predictable, guaranteed delays. L01 p5
LAN	Network of computers typically within 5 km, over copper or fiber, usually IEEE 802. L01 p6; L01 p7
WAN	Geographically long-distance network: fiber (incl. submarine cable), satellite or terrestrial microwave. L01 p6; L01 p10
Wireless network	Network over unguided media (radio, microwave); can be short or long distance; uses cells. L01 p6; L01 p12
IEEE 802	The IEEE family of LAN standards for the physical and data link layers (802.3 Ethernet, 802.5 token ring, 802.11 wireless, ...). L01 p8; L01 p9
Dark fiber	Unused fiber-optic cable. L01 p10
Starlink	SpaceX satellite constellation at 550 km bringing broadband to poorly served areas (775 satellites by October 2020). L01 p11

Media access control (MAC)	Coordination that lets a node gain access to a shared medium; ALOHA style or token style. L01 p13
Physical layer	Turns frames into electrical, optical or electromagnetic signals; specifies cables, connectors and signal levels. L01 p14
Analog signal	A signal that varies continuously; data rides on its amplitude, frequency or phase. L01 p15
Digital signal	A signal coded in binary levels — e.g. 1 = +5 V, 0 = 0 V. L01 p15
Modulator / demodulator	The modulator turns a digital signal into an analog one for transmission; the demodulator turns it back. L01 p17

Signals, Bandwidth, Noise & Data-Rate Limits

What a signal is, how sine waves combine into composite signals with a bandwidth, what impairs a signal on the way, and how fast a channel can possibly go (Nyquist and Shannon).

★ Nyquist & Shannon (slides 29–33)

★ dB and SNR

★ Latency: propagation vs transmission

★ Wavelength / period / phase

★ Summary slide 38

AT A GLANCE

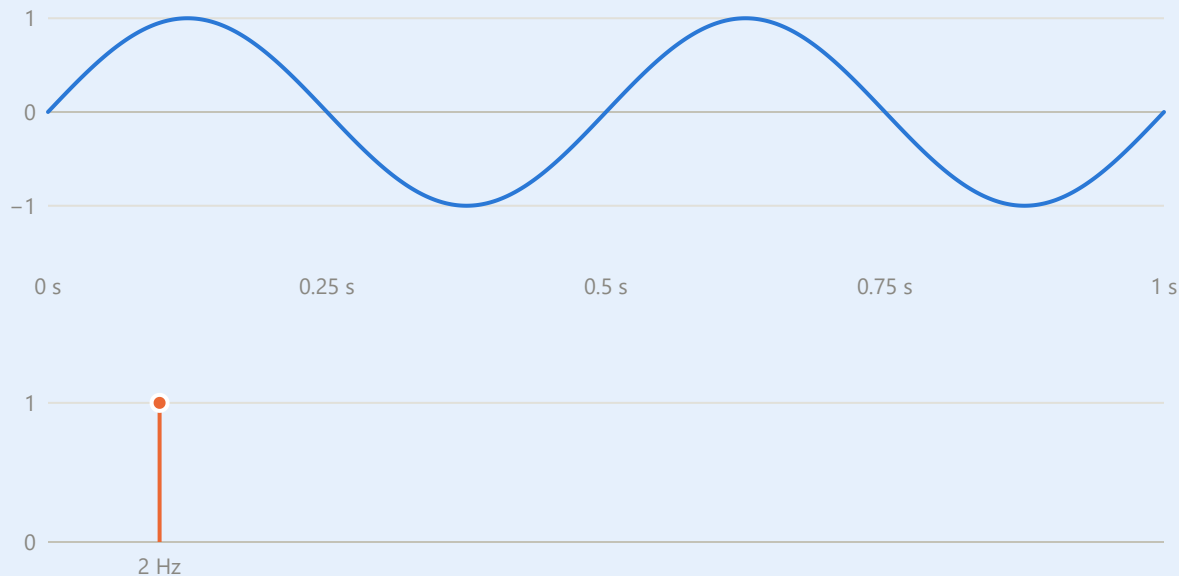
- Signals are **analog** (infinitely many levels) or **digital** (a limited number of levels), and **periodic** or **nonperiodic**. Data communications uses periodic analog and nonperiodic digital signals.
- A sine wave = **amplitude** (peak), **frequency** $f = 1/T$ in Hz, **phase** (position at time 0: 0° , 90° , 180° , 270°). **Wavelength** $\lambda = c/f$ ($c = 3 \times 10^8$ m/s): red light at 4×10^{14} Hz $\rightarrow$ 0.75 μ m.
- A **composite signal** is a sum of sine waves (Fourier); its **bandwidth** = highest – lowest frequency (5000 – 1000 = 4000 Hz). A digital signal is a composite analog signal with **infinite bandwidth**.
- **Baseband**: send the digital signal as is over a low-pass channel (the first harmonic needs $B = N/2$). **Broadband**: modulate it onto a bandpass channel.
- Impairments: **attenuation** ($\text{dB} = 10 \log_{10} P_2/P_1$; half power = -3 dB), **distortion**, **noise** (thermal, induced, crosstalk, impulse). **SNR** = signal/noise; $\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$ (10 mW / 1 μ W $\rightarrow$ 40 dB).
- **Nyquist** (noiseless): $N = 2B \log_2 L$. **Shannon** (noisy): $C = B \log_2(1 + \text{SNR})$. Use Shannon for the ceiling, then Nyquist for the levels (1 MHz, SNR 63 $\rightarrow$ 6 Mbps; use 4 Mbps $\rightarrow$ $L = 4$). **Latency** = propagation + transmission + queuing + processing.

WHAT THEY'LL ASK

- **Solving**: Shannon capacity, then a safe rate and the Nyquist levels (slide 33); Nyquist bit rate or levels.
- **Solving**: dB of a gain/loss (or a chain of them), SNR and SNR_{dB} from powers.
- **Solving**: propagation and transmission time (and total latency); wavelength, period, frequency, phase.
- **MCQ / Identification**: types of noise, attenuation vs distortion, baseband vs broadband, throughput vs bandwidth vs latency vs jitter, analog vs digital, periodic vs nonperiodic.
- **Essay**: Nyquist vs Shannon — when each applies and why more levels can't raise the rate forever; the summary points on slide 38.

💡 INTUITION FIRST

Notes, chords and how much road you need



Frequency $f = 2 \text{ Hz} \rightarrow$ period $T = 1/f = 0.5 \text{ s}$. Phase 0° .

A single sine wave occupies one frequency (bandwidth 0). Add harmonics to build a composite signal.

A sine wave is a single musical note: its **loudness** is the amplitude, its **pitch** the frequency, and *when* it starts is the phase. A real sound is a **chord** — many notes at once (a composite signal) — and the spread from its lowest to its highest note is its **bandwidth**. Sharp-edged digital pulses need many high “notes”, so they need a wide road.

Predict: You add the odd harmonics $3f$, $5f$, $7f$ and $9f$ to a sine wave of frequency f . What happens?

Answer: It approaches a square wave and its bandwidth grows

Fourier: a composite signal is a combination of sine waves. Adding odd harmonics with falling amplitudes sharpens the edges toward a square wave, and the bandwidth (highest – lowest frequency) grows. [L02 p11; L02 p12]

Predict: A channel has $\text{SNR}_{\text{dB}} = 0 \text{ dB}$. What is its Shannon capacity?

Answer: B bps (because $\text{SNR} = 1$ and $\log_2 2 = 1$)

0 dB means $\text{SNR} = 10^0 = 1$, so $C = B \log_2(1 + 1) = B$. Shannon needs the plain ratio — never put the dB value into the formula. [L02 p31]

Analog vs digital, periodic vs nonperiodic L02 pp3–5

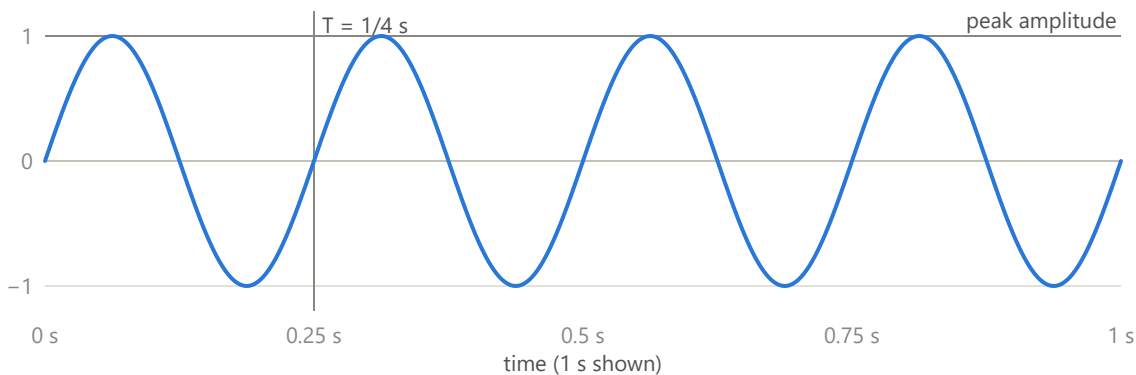
An **analog signal** has infinitely many levels of intensity over a period of time; a **digital signal** can have only a limited number of defined values. Signals are plotted with value (strength) on the vertical axis and time on the horizontal. L02 p4

A **periodic signal** repeats a pattern within a time frame called a **period**; one full pattern is a **cycle**. A **nonperiodic signal** changes without a repeating pattern. Both analog and digital signals can be periodic or nonperiodic. L02 p5

The sine wave: amplitude, frequency, period, phase L02 pp6–8

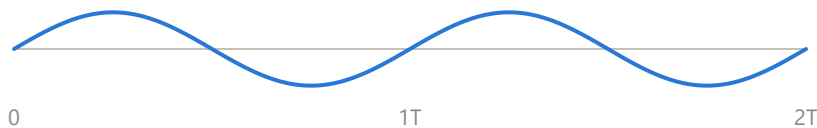
A sine wave is described by three parameters — **amplitude**, **frequency** and **phase**. L02 p6

- **(Peak) amplitude:** the absolute value of the highest intensity, proportional to the energy carried.
- **Frequency:** the number of periods in 1 s, in hertz ($\text{Hz} = \text{cycles per second}$). **Period:** the time in seconds to complete one cycle. $f = 1/T$, $T = 1/f$.
- **Phase** (phase shift): the position of the waveform relative to time 0, in degrees or radians.

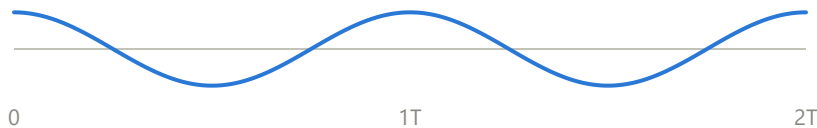


Slide 7: four cycles in 1 s $\rightarrow f = 4 \text{ Hz}$ and $T = 1/4 \text{ s}$. Peak amplitude is measured from the axis to the crest.

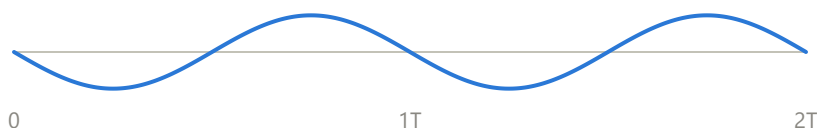
0° — starts at 0 going up



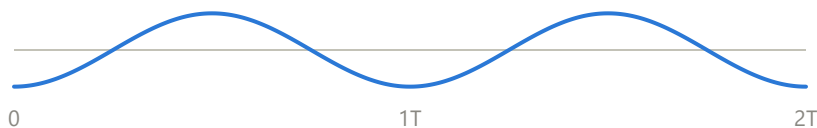
90° — starts at the peak



180° — starts at 0 going down



270° — starts at the trough



Slide 8: the same sine wave shifted by 0° , 90° , 180° and 270° .

WORKED EXAMPLE

Frequency and period (slide 7) L02 p7

A periodic signal completes **4 cycles in 1 s**. What are its frequency and its period?

1. Frequency is the number of periods in 1 s: $f = 4 \text{ cycles} / 1 \text{ s} = \mathbf{4 \text{ Hz}}$. L02 p6 L02 p7
2. Period and frequency are reciprocals ($f = 1/T$, $T = 1/f$): $T = 1/4 = \mathbf{0.25 \text{ s}}$. L02 p7

Frequency $f = 4 \text{ Hz}$ Period $T = 250 \text{ ms}$

Slide's answer: $f = 4 \text{ Hz}$, $T = 1/4 \text{ s}$

WORKED EXAMPLE

Phase from a time shift L02 p8

A sine wave has a frequency of **250 Hz**. A second wave of the same frequency **leads** it by **1 ms** (it starts 1 ms earlier). What is the phase of the second wave relative to the first, in degrees?

1. Period: $T = 1/f = 1/250 \text{ Hz} = 4 \text{ ms}$. L02 p7
2. A full period is 360° . Slide 8's pictures put the 90° wave a quarter period ahead of the 0° wave (it starts at the peak), 180° half a period and 270° three quarters, so phase = $360^\circ \times (\text{shift} / T)$. L02 p8
3. Phase = $360^\circ \times (1 \text{ ms} / 4 \text{ ms}) = 360^\circ \times 0.25 = \mathbf{90^\circ}$.

Phase shift = 90°

Wavelength L02 pp9–10

Wavelength binds the period or frequency of a sine wave to the propagation speed of the medium: $\lambda = \text{propagation speed} \times \text{period} = \mathbf{c / f}$, with $c = 3 \times 10^8 \text{ m/s}$ in free space (the speed of light). L02 p9

WORKED EXAMPLE

Wavelength of red light L02 p10

A sine wave has a frequency of **$4 \times 10^{14} \text{ Hz}$** and travels at a propagation speed of **$3 \times 10^8 \text{ m/s}$** (the speed of light in free space). What is its wavelength?

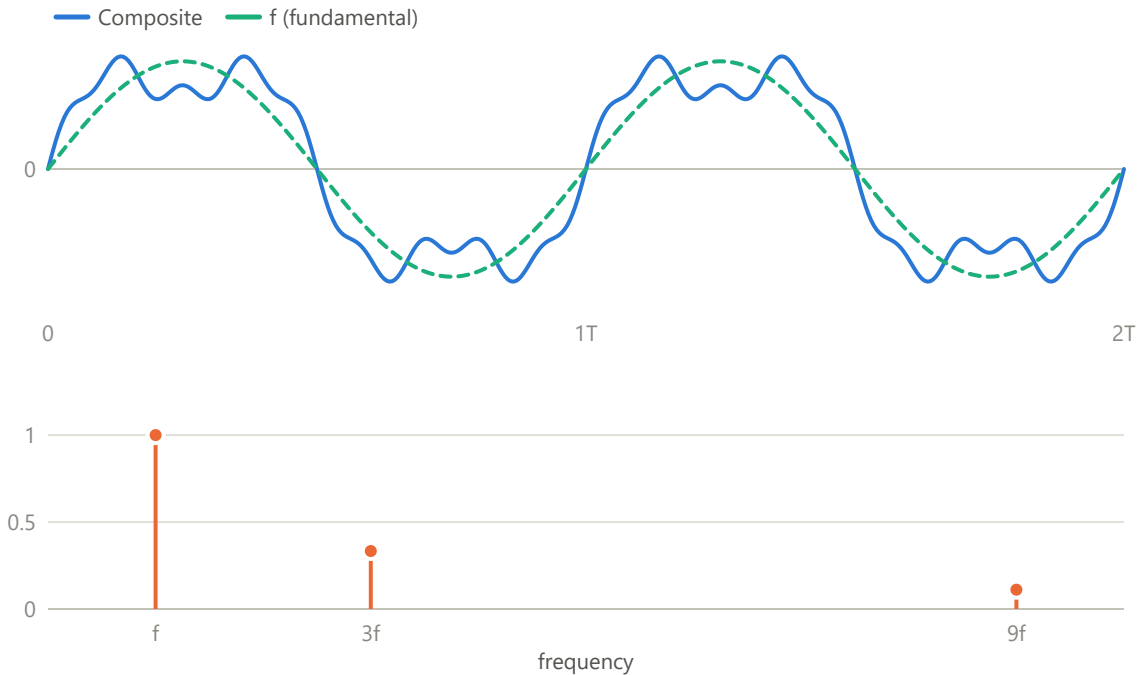
1. Wavelength binds the period or frequency to the propagation speed: $\lambda = \text{propagation speed} \times \text{period} = \text{propagation speed} / \text{frequency}$. L02 p9
2. $\lambda = v / f = 3 \times 10^8 / 4 \times 10^{14} = \mathbf{7.5 \times 10^{-7} \text{ m} = 0.75 \text{ }\mu\text{m}}$. L02 p10

Wavelength $\lambda = 750 \text{ nm}$

Slide's answer: $\lambda = 3 \times 10^8 / 4 \times 10^{14} = 0.75 \times 10^{-6} \text{ m} = 0.75 \text{ }\mu\text{m}$

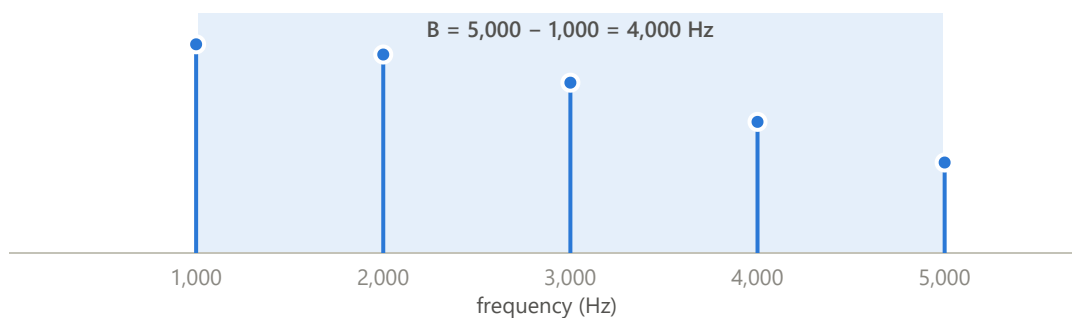
Composite signals and bandwidth L02 pp11–13

A **composite signal** is made of many simple sine waves with different frequencies, amplitudes and phases (Fourier); it can be periodic or nonperiodic. L02 p11 Slide 12's signal is built from f , $3f$ and $9f$; the main frequency is the **fundamental frequency**. L02 p12



Slide 12: a periodic composite signal built from sine waves at f , $3f$ and $9f$ with decreasing amplitudes; f is the fundamental frequency.

The **bandwidth** of a composite signal is the range of frequencies it contains — a **difference between two numbers**: highest – lowest. L02 p13



Slide 13: bandwidth = highest frequency – lowest frequency.

▲ SLIDE SAYS / CORRECT

Slide 11 Fourier showed this “in the early 1900s”

Correct Fourier’s work dates from the early 1800s (a slide-based question may still say 1900s).

🔧 WORKED EXAMPLE

Bandwidth of a periodic signal L02 p13

The frequency spectrum of a composite signal ranges from **1,000 Hz** to **5,000 Hz**. What is its bandwidth?

1. The bandwidth of a composite signal is the range of frequencies it contains: the difference between the highest and the lowest frequency. L02 p13
2. $B = f_{\max} - f_{\min} = 5,000 - 1,000 = \mathbf{4,000 \text{ Hz}}$.

Bandwidth $B = 4 \text{ kHz}$

Slide’s answer: $B = 5000 - 1000 = 4000 \text{ Hz}$

Digital signals: levels, bit rate, bit length L02 p14

A digital signal can have more than two levels: with **L levels**, each level carries **$\log_2 L$ bits**. Most digital signals are nonperiodic, so period and frequency don't describe them; we use:

- **Bit rate** — the number of bits sent in 1 s (bps). Slide 14: two levels, 8 signal elements in 1 s $\rightarrow$ 8 bps; four levels $\rightarrow$ 16 bps.
- **Bit length** — the distance one bit occupies on the medium. **Beyond slides** textbook formula: bit length = propagation speed $\times$ bit duration.

L02 p14

WORKED EXAMPLE

Four levels: bits per level and bit rate L02 p14

A digital signal sends **8 signal levels (elements) in 1 s**, and each element can take one of **4 levels**. How many bits does each level carry, and what is the bit rate?

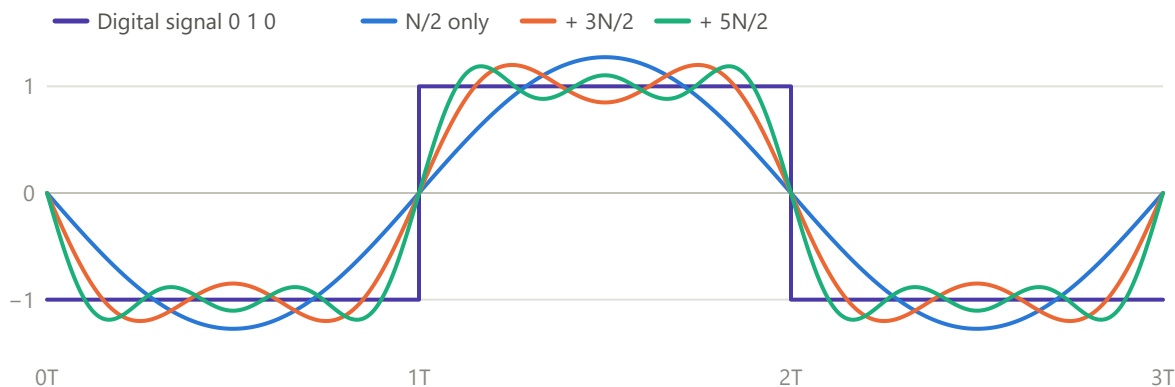
1. A signal with L levels needs $\log_2 L$ bits for each level: $\log_2 4 =$ **2 bits** per level. L02 p14
2. Bit rate = bits sent in 1 s = $8 \times 2 =$ **16 bps**. L02 p14

Bits per level = 2 Bit rate N = 16 bps

Slide's answer: 2 bits per level; 16 bits sent in 1 s $\rightarrow$ 16 bps

Baseband and broadband transmission L02 pp15–18

Baseband transmission sends a digital signal without changing it to an analog signal. It needs a wide-bandwidth (low-pass) channel — a dedicated medium whose bandwidth forms a single channel. L02 p16 In a low-pass channel with limited bandwidth the digital signal is approximated by an analog signal; how good the approximation is depends on the bandwidth available. L02 p17



Slide 17: in a low-pass channel the digital signal is approximated by an analog one; the first harmonic needs $B = N/2$, adding $3N/2$ and $5N/2$ needs $B = 5N/2$.

Broadband transmission changes the digital signal to an analog signal (modulation), which lets us use a **bandpass channel** — one whose bandwidth does not start from zero. L02 p18

SLIDE SAYS / CORRECT

Slides 16 and 18 The baseband channel is drawn from f_1 to f_2 , and the receiver's waveform on slide 18 is labelled "input"
Correct A baseband channel is low-pass (it starts at 0 Hz); the receiver side shows the output signal.

Transmission impairments: attenuation, distortion, noise L02 pp19–24

Attenuation is a loss of energy: a signal loses some energy overcoming the resistance of the medium. Amplifiers compensate for the loss. Engineers express gain or loss in **decibels**. L02 p20

$$\text{dB} = 10 \log_{10} (P_2 / P_1)$$

The decibel measures the relative strength of two signals, or of one signal at two points; it is **negative when the signal is attenuated** and **positive when it is amplified**. L02 p21



Decibels of cascaded stages add; convert the total back with $P_{\text{out}} = P_{\text{in}} \times 10^{(\text{dB}/10)}$.

Slide 20: the medium attenuates the signal (point 1 → point 2); an amplifier restores it (point 2 → point 3). Here: half the power lost (−3 dB), then doubled (+3 dB).

Distortion means the signal changes its form or shape. It happens in composite signals: the component frequencies travel at different speeds and arrive at different times. L02 p23

Noise corrupts the signal at the receiver: L02 p24

- **Thermal** — random motion of electrons in a wire.
- **Induced** — from motors and appliances: the source acts as a sending antenna and the medium as a receiving antenna.
- **Crosstalk** — signals jumping from one wire to an adjacent one.
- **Impulse** — a spike from power lines, lightning and the like.

⚠ SLIDE SAYS / CORRECT

Slide 22 “its power is reduced to one-half. This means that $P_2 = P_1$.”

Correct $P_2 = 0.5 P_1$ (the $\frac{1}{2}$ is missing), so $10 \log_{10} 0.5 = -3 \text{ dB}$.

🔧 WORKED EXAMPLE

Power halved: attenuation in dB L02 p22

A signal travels through a transmission medium and its power is reduced from $P_1 = 1 \text{ W}$ to $P_2 = 500 \text{ mW}$ (to one-half). What is the change in decibels? Is the signal attenuated or amplified?

1. Decibels compare two powers: $\text{dB} = 10 \log_{10}(P_2/P_1)$. L02 p21
2. $P_2/P_1 = 500 \text{ mW} / 1 \text{ W} = 0.5$.
3. $\text{dB} = 10 \log_{10}(0.5) = 10 \times (-0.301) = -3.01 \text{ dB}$ (the slide writes −3 dB). L02 p22
4. The decibel is negative if a signal is attenuated and positive if it is amplified: this one is **attenuated** (a loss). L02 p21

Change in decibels = **−3.01 dB**

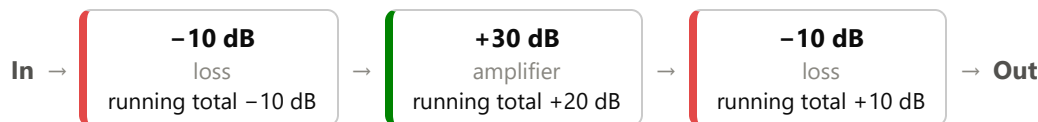
Slide's answer: $10 \log_{10} 0.5 = 10(-0.3) = -3 \text{ dB}$

WORKED EXAMPLE

A chain of losses and an amplifier L02 p21

A signal with a power of **2 mW** passes through a section with a loss of 10 dB, then an amplifier with a gain of 30 dB, and finally a section with a loss of 10 dB. Find the net change in dB and the output power.

1. Write every stage in dB: losses are negative, gains positive: $-10, +30, -10$ dB. L02 p21
2. Decibels of cascaded stages simply add: net $= -10 + 30 - 10 = \mathbf{10\text{ dB}}$ (a net gain). Forouzan
3. $P_{\text{out}}/P_{\text{in}} = 10^{10/10} = 10^1 = 10$, so $P_{\text{out}} = 2\text{ mW} \times 10 = \mathbf{20\text{ mW}}$. L02 p21



Decibels of cascaded stages add; convert the total back with $P_{\text{out}} = P_{\text{in}} \times 10^{(\text{dB}/10)}$.

Net change = 10 dB Output power = 20 mW

Signal-to-noise ratio (SNR) L02 pp25–26

SNR is the ratio of what is wanted (signal) to what is not wanted (noise): **SNR = average signal power / average noise power**, and **SNR_{dB} = 10 log₁₀ SNR**. A high SNR means the signal is less corrupted by noise; a low SNR means it is more corrupted.

L02 p25

WORKED EXAMPLE

SNR and SNR in dB L02 p26

The power of a signal is **10 mW** and the power of the noise is **1 μW**; what are the values of SNR and SNR_{dB}?

1. SNR = average signal power / average noise power, a plain ratio. L02 p25
2. $\text{SNR} = 10\text{ mW} / 1\text{ μW} = 0.01\text{ W} / 10^{-6}\text{ W} = \mathbf{10,000}$. L02 p26
3. $\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR} = 10 \log_{10} 10,000 = 10 \log_{10} 10^4 = 10 \times 4 = \mathbf{40\text{ dB}}$. L02 p25 L02 p26

SNR = 10,000 SNR_{dB} = 40 dB

Slide's answer: $\text{SNR} = 10,000\text{ μW} / 1\text{ μW} = 10,000$; $\text{SNR}_{\text{dB}} = 10 \log_{10} 10^4 = 40\text{ dB}$

Data-rate limits: Nyquist and Shannon L02 pp27–33

How fast we can send data (bps) depends on three things: the **bandwidth** available, the **number of signal levels**, and the **quality of the channel** (its noise). Two theoretical formulas apply. L02 p28 (Data rates use decimal prefixes: 1000 kbps = 1 Mbps.) L02 p27

Nyquist bit rate — noiseless channel

$$\text{BitRate} = 2 \times \text{bandwidth} \times \log_2 L$$

It assumes no noise. More levels raise the bit rate, but the receiver must still be able to tell the levels apart. L02 p29

Shannon capacity — noisy channel

$$\text{Capacity} = \text{bandwidth} \times \log_2(1 + \text{SNR})$$

There is no signal level in it: the number of levels does not change the capacity of the channel. SNR here is the plain ratio, not dB. L02 p31

Using both (slide 33)

Shannon gives the **upper limit**; choose a lower rate for better performance, then use Nyquist to find the number of signal levels.

L02 p33

WORKED EXAMPLE

Nyquist: noiseless 3000 Hz channel, 2 levels L02 p30

Consider a noiseless channel with a bandwidth of **3,000 Hz** transmitting a signal with **2 signal levels**. What is the theoretical maximum bit rate?

1. A noiseless channel follows Nyquist: bit rate = $2 \times \text{bandwidth} \times \log_2 L$. L02 p29
2. Bit rate = $2 \times 3,000 \times \log_2 2 = 2 \times 3,000 \times 1 = \mathbf{6,000 \text{ bps}}$. L02 p30

Bit rate $N = 6 \text{ kbps}$

Slide's answer: $2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$

WORKED EXAMPLE

Shannon: telephone line, $B = 3000 \text{ Hz}$, $\text{SNR} = 3162$ L02 p32

A channel has a bandwidth of **3,000 Hz** and a signal-to-noise ratio of **3,162** (a plain ratio, not dB). What is the theoretical highest bit rate (the capacity)?

1. A noisy channel follows Shannon: capacity = bandwidth $\times \log_2(1 + \text{SNR})$, with SNR as a plain ratio. L02 p31
2. $C = 3,000 \times \log_2(1 + 3,162) = 3,000 \times \log_2 3,163$.
3. $\log_2 3,163 \approx 11.627$, so $C = 3,000 \times 11.627 = \mathbf{34,881 \text{ bps}}$. L02 p32 The slide prints 34,860 bps because it rounds $\log_2 3,163$ to 11.62; the exact value is 34,881 bps.

Capacity $C = 34.881 \text{ kbps}$

Slide's answer: $3000 \times 11.62 = 34,860 \text{ bps}$ on the slide (it rounds $\log_2 3,163$ to 11.62); exact 34,881 bps

WORKED EXAMPLE

Nyquist and Shannon together L02 p33

We have a channel with a **1 MHz** bandwidth. The SNR for this channel is **63**. Find the upper limit of the bit rate. For better performance we then use a lower bit rate of **4 Mbps**: how many signal levels are needed?

1. The channel is noisy, so Shannon gives the upper limit: $C = B \times \log_2(1 + \text{SNR}) = 1 \text{ MHz} \times \log_2(1 + 63) = 1 \text{ MHz} \times \log_2 64 = 1 \text{ MHz} \times 6 = \mathbf{6 \text{ Mbps}}$. L02 p31 L02 p33
2. We may send at most C , and a lower rate is safer: $N = 4 \text{ Mbps} \leq 6 \text{ Mbps}$. L02 p33
3. Nyquist links the rate to the levels: $N = 2 \times B \times \log_2 L$, so $\log_2 L = 4,000,000 / (2 \times 1,000,000) = 2$. L02 p29
4. $L = 2^2 = \mathbf{4 \text{ signal levels}}$. L02 p33

Upper limit $C = 6 \text{ Mbps}$ Signal levels $L = 4$

Slide's answer: $C = 10^6 \log_2 64 = 6 \text{ Mbps}$ (upper limit); use 4 Mbps $\rightarrow 4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L \rightarrow L = 4$

Performance: bandwidth, throughput, latency, jitter L02 pp34–37

- **Bandwidth** can be in hertz or in bits per second — the higher the bandwidth in Hz, the higher the bit rate.
- **Throughput** is how fast we can actually send data: a link of bandwidth $B \text{ bps}$ carries only $T \text{ bps}$, with T always less than B .
- **Latency** (delay) is how long a whole message takes to arrive, from the moment its first bit is sent.
- **Jitter** is the difference in delay (latency) between packets.

Latency = propagation time + transmission time + queuing time + processing delay

Propagation time = distance / propagation speed. **Transmission time** = message size / bandwidth. L02 p35

■ Transmission 20 μ s ■ Propagation 50 ms

Latency = transmission + propagation = 50.02 ms. The largest share dominates the delay.

Slide 37: for a 2.5 KB e-mail over 12,000 km at 1 Gbps, propagation (50 ms) dwarfs transmission (0.020 ms).

✍ WORKED EXAMPLE

Propagation and transmission time of an e-mail L02 p37

What are the propagation time and the transmission time for a **2.5-KB (kilobyte)** message if the bandwidth of the network is **1 Gbps**? Assume that the distance between the sender and the receiver is **12,000 km** and that signals travel at **2.4×10^8 m/s**. (1 KB = 1000 bytes.)

1. Propagation time = distance / propagation speed = $(12,000 \times 1000 \text{ m}) / (2.4 \times 10^8 \text{ m/s}) = 50 \text{ ms}$. L02 p35

L02 p37

2. Message size in bits: 2,500 bytes $\times 8 = 20,000$ bits (the slides use KB = 1000 bytes).

3. Transmission time = message size / bandwidth = $20,000 \text{ bits} / 1 \text{ Gbps} = 20 \mu\text{s} (= 0.020 \text{ ms})$. L02 p35 L02 p37

■ Transmission 20 μ s ■ Propagation 50 ms

Latency = transmission + propagation = 50.02 ms. The largest share dominates the delay.

Propagation time $T_p = 50 \text{ ms}$ Transmission time $T_t = 20 \mu\text{s}$

Slide's answer: Propagation = $(12,000 \times 1000) / (2.4 \times 10^8) = 50 \text{ ms}$; transmission = $(2500 \times 8) / 10^9 = 0.020 \text{ ms}$

Summary (slide 38) L02 p38

★ LIKELY ASSESSED — THE SLIDE-38 SUMMARY

- In data communications we commonly use **periodic analog** signals and **nonperiodic digital** signals.
- A digital signal is a **composite analog signal with an infinite bandwidth**.
- For a **noiseless** channel, the **Nyquist** bit-rate formula gives the theoretical maximum bit rate.
- For a **noisy** channel, use the **Shannon** capacity to find the maximum bit rate.
- **Attenuation, distortion and noise** impair a signal.

Formulas

Frequency and period L02 p6; L02 p7

$$f = 1/T \cdot T = 1/f$$

f in Hz (cycles per second); T in seconds

Wavelength L02 p9

$$\lambda = c / f = c \times T$$

c = propagation speed (3×10^8 m/s in free space)

Phase from a time shift L02 p8; Forouzan**Beyond slides**

$$\text{phase} = 360^\circ \times t / T$$

$\frac{1}{4}$ period = 90° , $\frac{1}{2}$ period = 180°

Slide 8 only shows the four phase pictures (the 90° wave is a quarter period ahead); the formula is the textbook's.

Bandwidth L02 p13

$$B = f_{\text{high}} - f_{\text{low}}$$

range of frequencies in a composite signal

Bits per level L02 p14

$$\text{bits per level} = \log_2 L$$

L signal levels; bit rate = bits sent in 1 s

Decibel L02 p21; L02 p22

$$\text{dB} = 10 \log_{10}(P_2 / P_1)$$

negative = attenuated, positive = amplified; half power = -3 dB

Signal-to-noise ratio L02 p25

$$\text{SNR} = P_{\text{signal}} / P_{\text{noise}} \quad \text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$$

average powers; SNR is a plain ratio

Nyquist bit rate (noiseless) L02 p29

$$\text{BitRate} = 2 \times B \times \log_2 L$$

B = channel bandwidth (Hz); L = signal levels

Shannon capacity (noisy) L02 p31

$$C = B \times \log_2(1 + \text{SNR})$$

SNR as a plain ratio, NOT dB; no levels in the formula

Latency L02 p35

$$\text{Latency} = T_{\text{prop}} + T_{\text{trans}} + T_{\text{queue}} + T_{\text{proc}}$$

T_{prop} = distance / propagation speed; T_{trans} = message size / bandwidth

Bit length (textbook) Forouzan **Beyond slides**

$$\text{bit length} = \text{propagation speed} \times \text{bit duration}$$

$$\text{bit duration} = 1 / \text{bit rate}$$

⚠ COMMON TRAPS

- **Putting SNR in dB into Shannon's formula** → Convert first: $\text{SNR} = 10^{(\text{dB}/10)}$. 30 dB → 1000, not 30. L02 p31
- **Using Nyquist on a noisy channel to get the maximum rate** → Nyquist assumes no noise; the noisy-channel ceiling is Shannon's. Use Shannon first, then Nyquist for L. L02 p38
- **Thinking more levels always raise the rate** → Nyquist grows with $\log_2 L$, but the receiver must distinguish the levels; Shannon's limit has no L at all. L02 p29; L02 p31
- **Forgetting the sign in dB** → Loss → negative dB, gain → positive. Half power = -3 dB. L02 p21; L02 p22
- **Mixing bytes and bits in transmission time** → Message size in bits = bytes $\times$ 8 (2.5 KB = $2500 \times 8 = 20,000$ bits). L02 p37
- **Adding powers instead of dB** → Gains and losses of cascaded stages add in dB; powers multiply. L02 p21; Forouzan
- **Confusing attenuation with distortion** → Attenuation = loss of energy; distortion = change of shape (components arrive at different times). L02 p20; L02 p23

📌 MNEMONICS & MEMORY HOOKS

- Nyquist is **No**iseless and has **2** in front: $N = 2B \log_2 L$. Shannon is for **S**tatic (noise): $C = B \log_2(1 + \text{SNR})$.
- "3 dB = half": every -3 dB halves the power; every -10 dB divides it by 10.
- Latency = "**P**lease **T**ake **Q**uick **P**ictures": propagation + transmission + queuing + processing.
- Noise types — "**T**hermal **I**nduced **C**rosstalk **I**mpulse": TICl.
- Phase: 0° up from zero, 90° at the peak, 180° down through zero, 270° at the trough.

Blank-page recall

Close the notes. Answer each prompt from memory on paper, then check against the sections above.

1. Define amplitude, frequency, period, phase and wavelength; redo the red-light wavelength example.
2. What is a composite signal? Compute the bandwidth of the slide-13 signal.
3. Explain baseband vs broadband transmission and low-pass vs bandpass channels.
4. Write the dB and SNR formulas; redo slides 22 and 26.
5. Write the Nyquist and Shannon formulas and redo slides 30, 32 and 33.
6. Define throughput, latency and jitter; redo the slide-37 latency example.
7. Recite the five summary points of slide 38.

⚠ SLIDE ERRORS & CONTRADICTIONS

On the exam, the answer that follows the slide's intent is the safe one — but know the correct version.

Slide	Slide says	Correct / intended
L02 p11	Fourier showed this "in the early 1900s"	Early 1800s.
L02 p22	"This means that $P_2 = P_1$."	$P_2 = 0.5 P_1$ — the $\frac{1}{2}$ is missing; the answer is -3 dB.
L02 p20	Unit written "db"	dB (decibel).
L02 p7	The note writes the period as "t", and one "Amplitude" arrow spans crest to trough	Period $T = 1/f$; peak amplitude is measured from the axis to the crest.
L02 p26	Answer written as "40" with units "μw"	$\text{SNR}_{\text{dB}} = 40$ dB; the powers are in μW.
L02 p16	Baseband channel drawn from f_1 to f_2	A baseband (low-pass) channel starts at 0 Hz.

Glossary

Analog signal	A signal with infinitely many levels of intensity over a period of time. L02 p4
Digital signal	A signal that can take only a limited number of defined values. L02 p4
Periodic signal	A signal that repeats a pattern every period; one full pattern is a cycle. L02 p5
Nonperiodic signal	A signal that changes without a repeating pattern. L02 p5

Amplitude	The absolute value of a signal's highest intensity, proportional to the energy it carries. L02 p6
Frequency	The number of periods in 1 s, in hertz (Hz); $f = 1/T$. L02 p6
Period	The time, in seconds, a signal needs to complete one cycle; $T = 1/f$. L02 p6
Phase	The position of the waveform relative to time 0, in degrees or radians. L02 p6
Wavelength	$\lambda = \text{propagation speed} \times \text{period} = c/f$; binds frequency to the medium's propagation speed. L02 p9
Composite signal	A signal made of many simple sine waves with different frequencies, amplitudes and phases (Fourier). L02 p11
Fundamental frequency	The main (lowest) frequency of a composite signal. L02 p12
Bandwidth	The range of frequencies in a composite signal: highest minus lowest. L02 p13
Bit rate	The number of bits sent in 1 s (bps). L02 p14
Bit length	The distance one bit occupies on the transmission medium. L02 p14
Baseband transmission	Sending a digital signal over a (low-pass) channel without converting it to analog. L02 p16
Broadband transmission	Changing the digital signal to an analog one (modulation) so it can use a bandpass channel. L02 p18
Bandpass channel	A channel whose bandwidth does not start from zero. L02 p18
Attenuation	Loss of signal energy while overcoming the resistance of the medium; measured in dB. L02 p20
Decibel	$\text{dB} = 10 \log_{10}(P_2/P_1)$: relative strength of two signals, negative for a loss and positive for a gain. L02 p21
Distortion	A change in a signal's form or shape; components of a composite signal arrive at different times. L02 p23
Thermal noise	Noise from the random motion of electrons in a wire. L02 p24
Induced noise	Noise from motors and appliances acting as sending antennas, with the medium as the receiving antenna. L02 p24
Crosstalk	Noise from signals jumping from one wire to an adjacent wire. L02 p24
Impulse noise	A spike from power lines, lightning and the like. L02 p24
SNR	Average signal power / average noise power; $\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$. L02 p25
Nyquist bit rate	Maximum bit rate of a noiseless channel: $2 \times B \times \log_2 L$. L02 p29
Shannon capacity	Maximum bit rate of a noisy channel: $B \times \log_2(1 + \text{SNR})$; independent of the number of levels. L02 p31
Throughput	How fast data can actually be sent through a network; always less than the link's bandwidth. L02 p34
Latency	Time for an entire message to arrive: propagation + transmission + queuing + processing. L02 p34; L02 p35
Jitter	The difference in delay (latency) between packets. L02 p34

Digital Transmission I: Line Coding, Block Coding & Scrambling

How bits become a digital signal: the line-coding schemes and what makes one good (no DC, no baseline wander, self-clocking), then block coding and scrambling to fix the weak spots.

★ Draw NRZ-L / NRZ-I / Manchester / diff. Manchester / AMI

★ $S = c \cdot N / r$ and $B_{\min}$

★ 4B/5B + NRZ-I (Example 4)

★ B8ZS and HDB3

AT A GLANCE

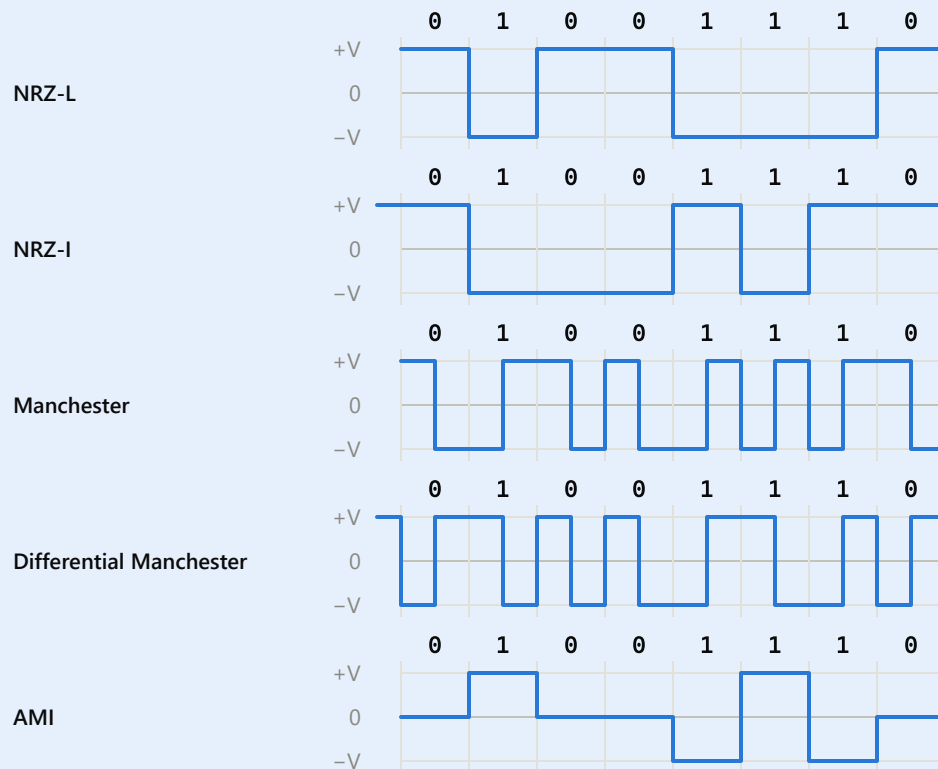
- **Line coding** turns data bits into signal levels. r = data elements per signal element; signal rate $S = c \times N \times 1/r$ (c : worst 1, best 0, average $1/2$). Goal: high data rate with a low baud rate. (Main Section 2)
- A good line code avoids **baseline wandering** and **DC components** (long runs of one level), is **self-synchronizing** (transitions to clock on), may **detect errors**, resists noise, and is not too complex. (Main Section 3)
- Slide-figure conventions: **NRZ-L** 0 = +V, 1 = -V · **NRZ-I** a 1 inverts · **RZ** 1 = +V→0, 0 = -V→0 · **Manchester** 0 = high→low, 1 = low→high · **Diff. Manchester** 0 = transition at the start · **AMI** 0 = 0 V, 1s alternate.
- Average baud / minimum bandwidth: NRZ-L, NRZ-I, AMI $N/2$; RZ, Manchester, diff. Manchester N ; 2B1Q $N/4$; MLT-3 $N/3$; 8B6T $3N/4$; 4D-PAM5 $N/8$.
- **Block coding** mB/nB adds redundancy: 4B/5B maps 4 bits to 5 (rate $\times 5/4$) so NRZ-I never sees long runs of 0s. 1 Mbps → 1.25 Mbps → NRZ-I needs 625 kHz, Manchester 1.25 MHz.
- **Scrambling** keeps the bandwidth and removes long zero runs on the fly: **B8ZS** 8 zeros → 000VB0VB; **HDB3** 4 zeros → 000V (odd count since last substitution) or B00V (even).

WHAT THEY'LL ASK

- **Solving (draw)**: encode a bit stream in NRZ-L, NRZ-I, RZ, Manchester, differential Manchester, AMI (and 2B1Q, MLT-3).
- **Solving**: average baud rate and minimum bandwidth for a scheme, with or without 4B/5B (Examples 1, 3, 4); clock drift (Example 2).
- **Solving**: apply B8ZS or HDB3 to a bit stream; encode with the 4B/5B table.
- **MCQ / Identification**: baseline wandering, DC component, self-synchronization, which scheme has which property, mBnL notation, violation/bipolar pulses.
- **Essay**: compare schemes on DC, baseline wandering and synchronization; explain why block coding or scrambling is used.

💡 INTUITION FIRST

Same bits, different shapes



Scheme	r	S (avg)	DC component	Self-sync
NRZ-L	1	$N/2$	yes (worse)	no (long 0s or 1s)
NRZ-I	1	$N/2$	yes	no (long 0s)
Manchester	$1/2$	N	no	yes (mid-bit transition)
Differential Manchester	$1/2$	N	no	yes (mid-bit transition)
AMI	1	$N/2$	no	no (long 0s)

Imagine sending bits by Morse-like light flashes to a friend counting seconds in their head. If you hold the light ON for a long time, they lose count (**no synchronization**) and their eyes adjust to the “new normal” (**baseline wandering**). Codes that **blink in the middle of every bit** (Manchester) keep the friend’s count in step — but blinking twice as often needs twice the bandwidth.

Predict: You send **00000000** with NRZ-L. What does the receiver see?

Answer: One long constant level — no transitions to stay in sync, and the baseline drifts

NRZ-L holds one level for the whole run, so there are no transitions (no self-synchronization) and the running average drifts (baseline wandering). [L03 p11; L03 p24]

Predict: Which scheme has a transition in the **middle of every bit**, no matter what the data is?

Answer: Manchester

Manchester (and differential Manchester) always change level mid-bit — that transition is the clock. The cost: twice the bandwidth of NRZ. [L03 p28; L03 p30]

Line coding: data elements vs signal elements L03 pp3-8

Line coding converts a string of 1s and 0s (digital data) into a sequence of signals — e.g. a high voltage (+V) for “1” and a low voltage (0 or –V) for “0”. The sender’s encoder makes the signal; the receiver’s decoder recovers the bits. L03 p3 L03 p4

A **data element** (data symbol) is one bit or a group of bits — 1, 0 or 11, 10, 01, ...; it is coded into one or more **signal elements** ($1 \rightarrow +V$, or $1 \rightarrow +V$ then $-V$). The ratio r = data elements carried per signal element. L03 p5

Slide 7 case	r	Example
(a) one data element per signal element	1	NRZ-type codes
(b) one data element per two signal elements	$\frac{1}{2}$	Manchester, RZ (two halves per bit)
(c) two data elements per signal element	2	11 01 11 $\rightarrow$ three signal elements (multilevel codes; 2B1Q is the slides’ $r = 2$ code)
(d) four data elements per three signal elements	$\frac{4}{3}$	8B6T-style multilevel codes

Data rate N = bits per second (bit rate). **Signal rate S** = signal elements per second, in **baud** (also called the modulation rate). The goal is to increase the data rate while reducing the baud rate. L03 p6

$$S = c \times N \times 1/r \text{ (baud)}$$

c is the **case factor**: worst case 1, best case 0, and the **average case** $\frac{1}{2}$, which is what the examples use. L03 p8

★ HIGH YIELD

The handwritten marks call line coding “Main Section 1”, data rate vs signal rate “Main Section 2” and synchronization “Main Section 3”, with the examples as sub-sections.

✍ WORKED EXAMPLE

Example 1: average baud rate L03 p9

A signal is carrying data in which one data element is encoded as one signal element ($r = 1$). If the bit rate is **100 kbps**, what is the average value of the baud rate if c is between 0 and 1?

1. The baud (signal) rate is $S = c \times N \times 1/r$, where c is the case factor (worst case 1, best case 0). L03 p8
2. For the average case we take $c = 1/2$. L03 p9 L03 p25
3. $r = 1$, so $1/r = 1$. L03 p7
4. $S = \frac{1}{2} \times 100,000 \times 1 = 50 \text{ kbaud}$.

Average baud rate $S = 50 \text{ kbaud}$

Slide’s answer: $S = \frac{1}{2} \times 100,000 \times 1 = 50 \text{ kbaud}$

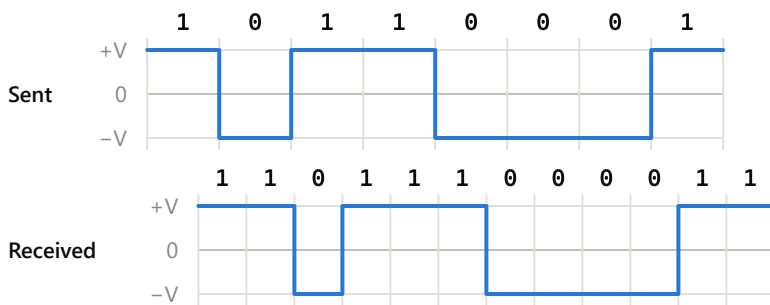
Bandwidth of a digital signal L03 p10

The actual bandwidth of a digital signal is infinite, but its **effective bandwidth is finite**. The minimum bandwidth follows the signal rate: $B_{\min} = S$ (Example 3), and the Nyquist limit still applies: $N_{\max} = 2 \times B \times \log_2 L$ for a noiseless channel (L02).

L03 p10 L03 p25

What makes a good line code L03 pp11–17

- **Baseline wandering** — the receiver computes a running average of the received signal power, the **baseline**. A long string of 0s or 1s makes the baseline drift and decoding hard. A good code prevents long runs of one amplitude. L03 p11
- **DC components** — when the voltage stays constant for a long time, low frequencies increase. Most channels are bandpass and cannot pass them, so the DC component must be removed. L03 p12
- **Self-synchronization** — sender and receiver clocks must use the same bit interval; a faster or slower receiver clock misreads the stream. Transitions in the signal let the receiver keep its clock aligned. L03 p13
- **Error detection** — some codes make certain transitions illegal, so the receiver notices a symbol error. L03 p15
- **Noise and interference immunity** — some codes make the signal “immune” to noise, which is stronger than just detecting errors. L03 p16
- **Complexity** — the more robust the code, the more complex it is, often paid for in baud rate or bandwidth. L03 p17



Slide 13: the receiver's faster clock samples the same waveform 12 times instead of 8, so 10110001 is read as 110111000011.

WORKED EXAMPLE

Example 2: a receiver clock 0.1% fast L03 p14

In a digital transmission, the receiver clock is **0.1% faster** than the sender clock. How many extra bits per second does the receiver receive if the data rate is **1 kbps**? How many if the data rate is **1 Mbps**?

1. The clocks must run at the same bit interval; if the receiver clock is faster or slower, it misreads the stream. L03 p13
2. The receiver reads $N \times (1 + 0.1/100)$ bits per second, so the extra bits are $N \times 0.1/100$. L03 p14
3. At 1 kbps: $1,000 \times 0.1/100 = \mathbf{1 \text{ bps}}$ extra; the receiver gets 1,001 bps instead of 1,000 bps.
4. At 1 Mbps: $1,000,000 \times 0.1/100 = \mathbf{1 \text{ kbps}}$ extra; the receiver gets 1,001,000 bps instead of 1,000,000 bps.

At 1 kbps = **1 bps** At 1 Mbps = **1 kbps**

Slide's answer: 1001 bps instead of 1000 (1 extra bps); 1,001,000 bps instead of 1,000,000 (1000 extra bps)

Unipolar and polar NRZ (NRZ-L, NRZ-I) L03 pp18–25

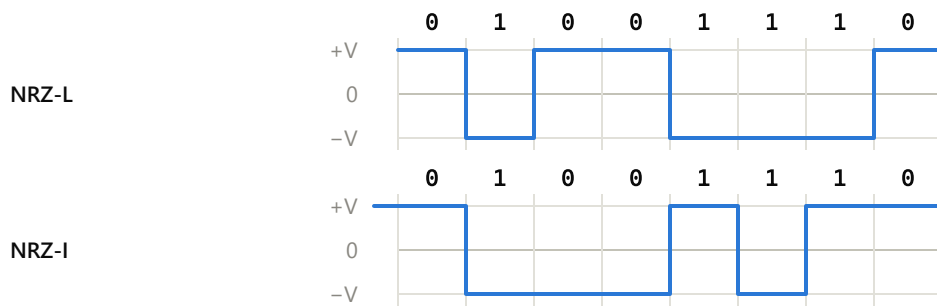
The schemes fall into five families (slide 19): **unipolar** (NRZ), **polar** (NRZ, RZ, biphasic: Manchester and differential Manchester), **bipolar** (AMI, pseudoternary), **multilevel** (2B1Q, 8B6T, 4D-PAM5) and **multitransition** (MLT-3). L03 p19

Unipolar NRZ — all levels on one side of the axis: 1 = +V, 0 = 0 V, and the level does not return to zero during a bit. Prone to baseline wandering and DC; no synchronization or error detection; simple but **costly in power** (normalized power $\frac{1}{2}V^2$).

L03 p20 L03 p21

Polar NRZ uses +V and -V:

- **NRZ-L** (level): the voltage **level** decides the bit. Slide-figure convention: **0 = +V, 1 = -V**.
- **NRZ-I** (inversion): a **1 inverts** the level at the start of the bit; a 0 keeps it. The figure starts at +V.



Slide 23: 01001110 in NRZ-L (0 = +V) and NRZ-I (a 1 inverts; starts at +V).

- Both: average signal rate **$N/2$** , DC component and baseline wandering (**worse for NRZ-L**), no self-synchronization (NRZ-I loses sync only on long 0s), no error detection, simple. L03 p24

⚠ SLIDE SAYS / CORRECT

Slide 22 (text) "+V for 1 and -V for 0"

Correct The NRZ-L figure on slide 23 uses 0 = +V and 1 = -V. Either polarity is a convention — state which one you use when drawing.

🔧 WORKED EXAMPLE

Example 3: NRZ-I at 1 Mbps L03 p25

A system is using **NRZ-I** to transfer **1 Mbps** data. What are the average signal rate and the minimum bandwidth?

- NRZ-I has $r = 1$ (one signal element per bit). L03 p23
- Average case ($c = 1/2$): $S = c \times N \times 1/r = 1/2 \times 1 \text{ Mbps} \times 1 = \mathbf{500 \text{ kbaud}}$. L03 p8 L03 p46
- The minimum bandwidth equals the average signal rate: $B_{\min} = S = \mathbf{500 \text{ kHz}}$. L03 p25

Average baud rate $S = \mathbf{500 \text{ kbaud}}$ Minimum bandwidth = **500 kHz**

Slide's answer: $S = 1/2 \times 1 \text{ Mbps} \times 1 = 500 \text{ kbaud}$; $B_{\min} = S = 500 \text{ kHz}$

WORKED EXAMPLE

Draw NRZ-L and NRZ-I (slide 23) L03 p23

Draw the bit stream **01001110** as a waveform in each of the line codes below (fill every cell of the grid). Use the slide conventions:

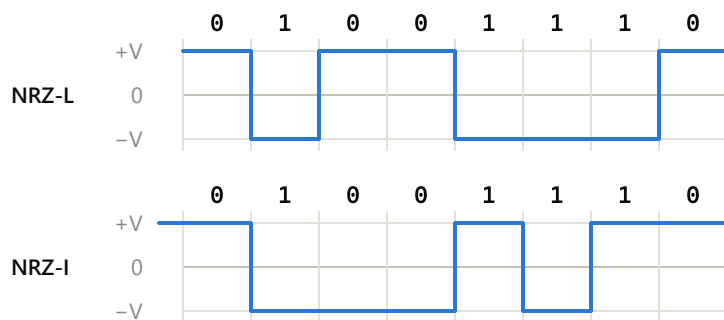
- NRZ-L (slide figure): 0 = +V, 1 = -V.
- NRZ-I: a 1 inverts the level at the start of the bit, a 0 does not; assume the level before the first bit is +V.

1. **NRZ-L** — 0 = +V, 1 = -V. L03 p23

2. Bit by bit: 0→+, 1→-, 0→+, 0→+, 1→-, 1→-, 1→-, 0→+ — the waveform is + - + + - - - +.

3. **NRZ-I** — a 1 inverts the level at the start of the bit, a 0 does not; assume the level before the first bit is +V. L03 p23

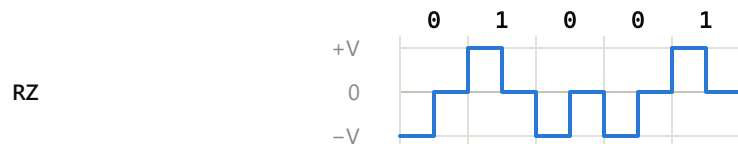
4. Bit by bit: 0→+, 1→-, 0→-, 0→-, 1→+, 1→-, 1→+, 0→+ — the waveform is + - - - + - + +.



Slide's answer: NRZ-L + - + + - - - + · NRZ-I + - - - + - + +

Polar RZ and biphase (Manchester, differential Manchester) L03 pp26-30

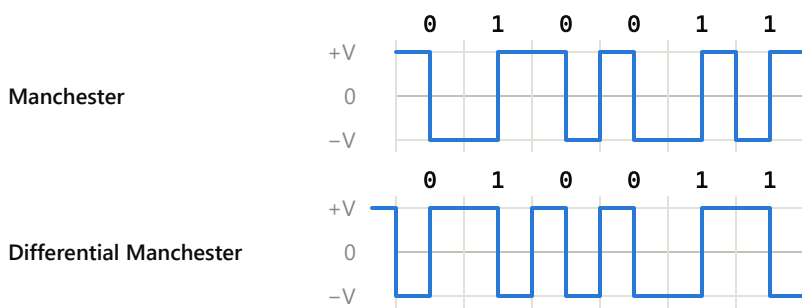
RZ (return to zero) uses three values +, 0, -. Every bit has a transition in the middle — **1 = +V then 0**, **0 = -V then 0**. Two transitions per bit → wider bandwidth; no DC or baseline wandering; self-synchronizing; more complex (three levels); no error detection. $r = \frac{1}{2}$, $S = N$. L03 p26 L03 p27



Slide 27: 01001 in RZ.

Manchester = NRZ-L + RZ: every bit has a level transition in the middle, using only two levels. Slide legend: **0 = high → low**, **1 = low → high**. L03 p28 L03 p29

Differential Manchester = NRZ-I + RZ: a transition in the middle of every bit, and the value is decided at the **start** of the bit — a **0 changes level** at the start, a 1 does not. L03 p28



Slide 29: 010011 in Manchester and differential Manchester (level before the first bit: +).

- The mid-bit transition is used for **synchronization**; minimum bandwidth is **2 × that of NRZ** ($S = N$); **no DC** and no baseline wandering; no error detection. **L03 p30**

△ SLIDE SAYS / CORRECT

Slide 5 (text) "1 → +V and -V, 0 → -V and +V" (1 = high→low)

Correct The Manchester legend on slide 29 is 0 = high→low, 1 = low→high. Follow the figure, and state your convention.

✍ WORKED EXAMPLE

Draw Manchester and differential Manchester (slide 29) **L03 p29**

Draw the bit stream **010011** as a waveform in each of the line codes below (fill every cell of the grid). Use the slide conventions:

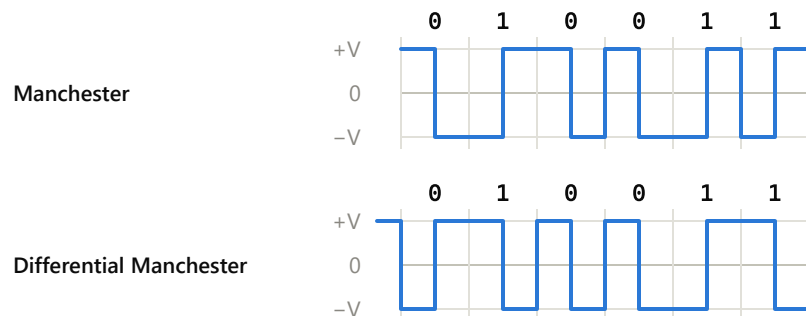
- Manchester: 0 = high→low, 1 = low→high (a transition in the middle of every bit).
- Differential Manchester: always a transition in the middle of the bit; a 0 also has a transition at the start of the bit, a 1 does not; assume the level before the first bit is +V.

1. **Manchester** — 0 = high→low, 1 = low→high (a transition in the middle of every bit). **L03 p29**

2. Bit by bit: 0→+-, 1→-+, 0→+-, 0→+-, 1→-+, 1→-+ — the waveform is (+-)(-+)(+-)(+-)(-+)(-+).

3. **Differential Manchester** — always a transition in the middle of the bit; a 0 also has a transition at the start of the bit, a 1 does not; assume the level before the first bit is +V. **L03 p29**

4. Bit by bit: 0→-+, 1→+-, 0→-+, 0→-+, 1→-+, 1→-+ — the waveform is (-+)(+-)(+-)(+-)(-+)(-+).



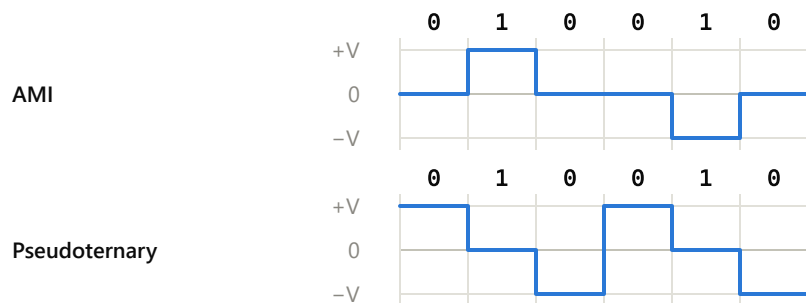
Slide's answer: Manchester (+-)(-+)(+-)(+-)(-+)(-+) · diff. Manchester (-+)(+-)(+-)(+-)(-+)(-+)

Bipolar: AML and pseudoternary **L03 pp31-33**

Three voltage levels +, 0, -, but no return to zero within a bit. One symbol is 0 V and the other **alternates** between + and -.

L03 p31

- **AMI** (alternate mark inversion): **0 = 0 V, 1s alternate +V, -V** (the first 1 is +V in the slide figure).
- **Pseudoternary**: the reverse — **1 = 0 V, 0s alternate**. **Low priority** the handwritten note on slide 32 says “not included”.



Slide 32: 010010 in AMI and pseudoternary.

- A better alternative to NRZ: **no DC component or baseline wandering**; but **no self-synchronization** for long runs of 0s (no transitions), and no error detection. $S = N/2$. **L03 p33**

Multilevel schemes: mBnL, 2B1Q, 8B6T, 4D-PAM5 **L03 pp34–42**

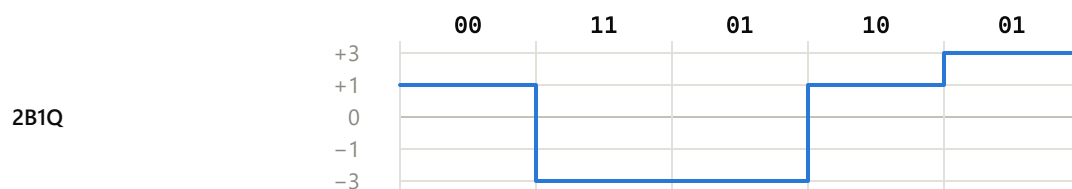
Multilevel schemes pack more data bits into each symbol. m data bits give 2^m data patterns; n signal elements with L levels give L^n signal patterns:

- $2^m > L^n$: not enough signal patterns — impossible.
- $2^m = L^n$: an exact mapping, no redundancy.
- $2^m < L^n$: extra patterns — choose the most distinct ones for noise immunity, and the unused ones signal errors.

So an **mBnL** code needs $2^m \leq L^n$. In the name, m = binary pattern length, B = binary data, n = signal pattern length, and L is written B (2 levels), T (3, ternary) or Q (4, quaternary). **L03 p36** **L03 p37**

2B1Q (2 bits → 1 quaternary symbol)

Next bits	Previous level +	Previous level -
00	+1	-1
01	+3	-3
10	-1	+1
11	-3	+3



Slide 38: 00 11 01 10 01 → +1, -3, -3, +1, +3 (positive starting level). Average signal rate $N/4$.

⚠ SLIDE SAYS / CORRECT

Slide 38 The box says $r = 1/2$

Correct $r = 2$ (two bits per level); that is the only value consistent with $S_{avg} = N/4$.

Redundancy: 2B1Q has none, so a DC component is present. Codes with redundancy can use only “0- or +-weighted” patterns and invert any pattern that would create DC — e.g. +00++- is sent as -00--+; the receiver inverts it back. L03 p39

8B6T maps 8 bits to 6 ternary symbols (slide examples: 00010001 → -0-0+; 01010011 → -+--+0; 01010000 → +--+0+, sent inverted as -++-0- to keep the DC balance). L03 p40

Multiple channels (xD-YYYz): split the signal across x links at once, lowering the signal rate per link; YYYz = z levels of a modulation type such as PAM. **4D-PAM5** sends 00011110 as -2, +1, +2, -1 on four wires: 1 Gbps → 250 Mbps per wire at 125 Mbaud. L03 p41 L03 p42

WORKED EXAMPLE

Is 8B6T a valid mBnL code? L03 pp34–37

A designer proposes the **8B6T** code: every pattern of $m = 8$ data bits is replaced by a pattern of $n = 6$ signal elements, each with $L = 3$ levels (ternary). How many data patterns and signal patterns are there? Is the code valid? How many signal patterns go unused (answer 0 if the code is not valid)?

1. mBnL notation: m data bits → n signal elements with L levels; the letter gives L ($B = 2, T = 3, Q = 4$), so 8B6T has $m = 8$, $n = 6$, $L = 3$. L03 p37
2. Data patterns: $2^m = 2^8 = 256$. Signal patterns: $L^n = 3^6 = 729$. L03 p34 L03 p35
3. A valid mBnL code needs $2^m \leq L^n$. L03 p36
4. 256 data patterns against 729 signal patterns: the data patterns are fewer than the signal patterns ($2^m < L^n$): the spare signal patterns can be chosen to be more distinct, giving better noise immunity and error detection → the code is **valid**. L03 p35
5. Unused patterns = $L^n - 2^m = 729 - 256 = 473$.

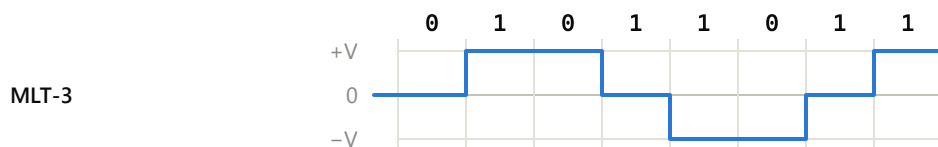
Data patterns $2^m = 256$ Signal patterns $L^n = 729$ Is the code valid? = Yes, it is valid

Unused signal patterns = 473

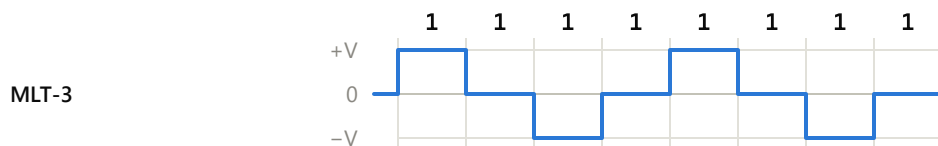
Slide's answer: $2^8 = 256 \leq 3^6 = 729 \rightarrow$ valid, with 473 unused patterns

Multitransition: MLT-3 L03 pp43–45

MLT-3 cycles through three levels. Rules: a **0 keeps the level**; a **1 moves** — from +V or -V to 0, and from 0 to the opposite of the last nonzero level. The slide trace starts at 0 V with its first 1 going to +V. L03 p44



Slide 44 (a): typical case 01011011 → 0, +, +, 0, -, -, 0, +.



Slide 44 (b): worst case 11111111 → +, 0, -, 0, +, 0, -, 0 — periodic, one period every 4 bits.

Its signal rate is the same as NRZ-I, but the worst-case pattern 1111 is periodic and behaves like an analog signal at $\frac{1}{4}$ of the bit rate. L03 p45 The summary table lists $B = N/3$. L03 p46

Summary of line coding schemes L03 p46

Category	Scheme	Bandwidth (avg)	Characteristics
Unipolar	NRZ	$B = N/2$	Costly, no self-synchronization if long 0s or 1s, DC
Polar	NRZ-L	$B = N/2$	No self-synchronization if long 0s or 1s, DC
Polar	NRZ-I	$B = N/2$	No self-synchronization for long 0s, DC
Polar	Biphase	$B = N$	Self-synchronization, no DC, high bandwidth
Bipolar	AMI	$B = N/2$	No self-synchronization for long 0s (no DC — slide 33)
Multilevel	2B1Q	$B = N/4$	No self-synchronization for long same double bits
Multilevel	8B6T	$B = 3N/4$	Self-synchronization, no DC
Multilevel	4D-PAM5	$B = N/8$	Self-synchronization, no DC
Multitransition	MLT-3	$B = N/3$	No self-synchronization for long 0s

▲ SLIDE SAYS / CORRECT

Slide 46 NRZ-L, NRZ-I and biphase are labelled “Unipolar”; AMI is marked “DC”; MLT-3’s category is “Multiline”

Correct They are polar codes; AMI has no DC component (slide 33); the category is multitransition.

8B6T: the slide table gives $B = 3N/4$ — use it for questions on this table. The general formula with $c = 1/2$ and $r = 4/3$ would give $S = 3N/8$; the textbook still puts 8B6T’s minimum bandwidth at about $6N/8 = 3N/4$.

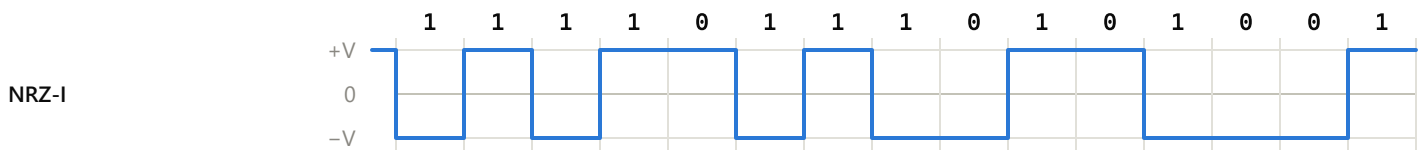
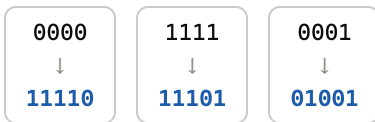
Block coding: 4B/5B and 8B/10B L03 pp47–56

Error detection and synchronization both need **redundancy**. Block coding adds it in three steps — **division** into m-bit groups, **substitution** with n-bit groups, **combination** into one stream. It is written **mB/nB** (with a slash, unlike multilevel mBnL) and avoids bit patterns that would cause DC or poor synchronization in the line code. L03 p48 L03 p49

4B/5B is used with **NRZ-I**: NRZ-I fails on long runs of 0s, and no code word in the slide-51 table starts with more than one 0 or ends with more than two, so the coded stream never has more than three 0s in a row. L03 p50

Data	Code	Data	Code
0000	11110	1000	10010
0001	01001	1001	10011
0010	10100	1010	10110
0011	10101	1011	10111
0100	01010	1100	11010
0101	01011	1101	11011
0110	01110	1110	11100
0111	01111	1111	11101

Control codes: Q (Quiet) = 00000 · I (Idle) = 11111 · H (Halt) = 00100 · J (Start delimiter) = 11000 · K (Start delimiter) = 10001 · T (End delimiter) = 01101 · S (Set) = 11001 · R (Reset) = 00111. L03 p51



Division into 4-bit groups → substitution with 5-bit codes → combination, then NRZ-I line coding (slide 50). The long run of 0s in 0000 becomes 11110.

Redundancy: $2^4 = 16$ data words, $2^5 = 32$ code words → $32 - 16 = 16$ extra words, some used for control and signalling.

L03 p53

⚠ SLIDE SAYS / CORRECT

Slide 53 "24 combinations ... $2^5 = 32$... $32 - 16 = 16$ "

Correct $2^4 = 16$, $2^5 = 32$, and $32 - 16 = 16$ (lost superscripts and a typo).

8B/10B = a 5B/6B encoder plus a 3B/4B encoder and a **disparity controller**; more redundant bits let it pick code words that prevent long runs of one voltage (DC). L03 p55 L03 p56

🔧 WORKED EXAMPLE

Example 4: 4B/5B with NRZ-I L03 p54

We need to send data at **1 Mbps**. Using **4B/5B block coding followed by NRZ-I** line coding, find the bit rate after block coding, the average signal (baud) rate and the minimum required bandwidth.

1. 4B/5B block coding replaces every 4 bits by 5 bits, so the bit rate rises to $N' = N \times 5/4 = 1 \text{ Mbps} \times 5/4 = \mathbf{1.25 \text{ Mbps}}$.

L03 p48 L03 p54

2. NRZ-I has $r = 1$ (one signal element per bit). L03 p23

3. Average case ($c = 1/2$): $S = c \times N \times 1/r = 1/2 \times 1.25 \text{ Mbps} \times 1 = \mathbf{625 \text{ kbaud}}$. L03 p8 L03 p46

4. The minimum bandwidth equals the average signal rate: $B_{\min} = S = \mathbf{625 \text{ kHz}}$. L03 p25

5. Trade-off: NRZ-I needs the lower bandwidth ($N'/2$) but has a DC-component problem; Manchester would need $N' = 1.25 \text{ MHz}$ but has no DC component. L03 p54

Bit rate after 4B/5B = **1.25 Mbps** Average baud rate $S = \mathbf{625 \text{ kbaud}}$ Minimum bandwidth = **625 kHz**

Slide's answer: 1 Mbps → 1.25 Mbps; NRZ-I minimum bandwidth $N/2 = 625 \text{ kHz}$

WORKED EXAMPLE

Example 4 (other choice): 4B/5B with Manchester L03 p54

We need to send data at **1 Mbps**. Using **4B/5B block coding followed by Manchester** line coding, find the bit rate after block coding, the average signal (baud) rate and the minimum required bandwidth.

1. 4B/5B block coding replaces every 4 bits by 5 bits, so the bit rate rises to $N' = N \times 5/4 = 1 \text{ Mbps} \times 5/4 = \mathbf{1.25 \text{ Mbps}}$.

L03 p48 L03 p54

2. Manchester has $r = 1/2$ (two signal elements per bit). L03 p29

3. Average case ($c = 1/2$): $S = c \times N \times 1/r = 1/2 \times 1.25 \text{ Mbps} \times 2 = \mathbf{1.25 \text{ Mbaud}}$. L03 p8 L03 p46

4. The minimum bandwidth equals the average signal rate: $B_{\min} = S = \mathbf{1.25 \text{ MHz}}$. L03 p25

5. Trade-off: Manchester needs twice the bandwidth of NRZ-I (N' against $N'/2$) but has no DC component. L03 p54

Bit rate after 4B/5B = **1.25 Mbps** Average baud rate $S = \mathbf{1.25 \text{ Mbaud}}$ Minimum bandwidth = **1.25 MHz**

Slide's answer: Manchester needs 1.25 MHz — more bandwidth, but no DC problem

WORKED EXAMPLE

Encode with the 4B/5B table L03 p51

Using the 4B/5B table, encode the data stream **0000 1111 0001** for transmission. If the data rate is **1 Mbps**, what is the bit rate on the line after block coding?

1. Division: split the stream into 4-bit groups **0000 1111 0001**. L03 p48 L03 p49

2. Substitution: replace each group using the table (data sequence → encoded sequence): **0000** → **11110**, **1111** → **11101**, **0001** → **01001**. L03 p51 L03 p52

3. Combination: join the 5-bit words: **11110 11101 01001** (never more than three 0s in a row). L03 p49

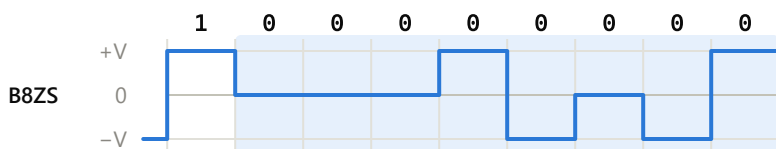
4. Five bits replace four, so the rate becomes $N \times 5/4 = 1 \text{ Mbps} \times 5/4 = \mathbf{1.25 \text{ Mbps}}$. L03 p54

4B/5B stream = **111101110101001** Bit rate after 4B/5B = **1.25 Mbps**

Scrambling: B8ZS and HDB3 L03 pp57–61

The best code adds no bandwidth for synchronization and has no DC. **Scrambling** creates such a stream on the fly, during encoding: it replaces “unfriendly” runs of bits with a **violation code** that is easy to recognize. It is used with AMI (“modified AMI”). L03 p57 L03 p58

- **B8ZS** replaces **eight consecutive zeros with 000VB0VB**. **V** = violation (breaks the AMI rule: same polarity as the previous pulse); **B** = bipolar (follows the AMI rule). L03 p59



Pulse before the stream: -. Substitutions: bits 2–9 → 000VB0VB. Marks: V at bit 5, B at bit 6, V at bit 8, B at bit 9.

Slide 59 (a): previous level positive → 000+–0–+.

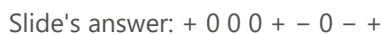


-

Slide 61: 11 0000 1 0000 0000 0 $\rightarrow$ + - | +00+ | - | 000- | +00+ | 0.

Correct The receiver decodes — it should say decoding.

1. Scrambling works on top of AMI: 0 = 0 V and the 1s alternate in polarity. The last pulse before the stream was negative (−V), so the next 1 is +. **L03 p58**
2. B8ZS replaces eight consecutive 0s with **000VB0VB**: V = a violation (the same polarity as the previous pulse), B = a normal bipolar pulse (the opposite polarity). **L03 p59**
3. Bits 2–9 are eight 0s and the pulse before them is positive (+V): V = +, B = −, 0, V = − (the same as the B before it), B = + → **0 0 0 + − 0 − +**. **L03 p59**
4. The transmitted pulses are **+ 0 0 0 + − 0 − +** — no run of eight zeros is left.

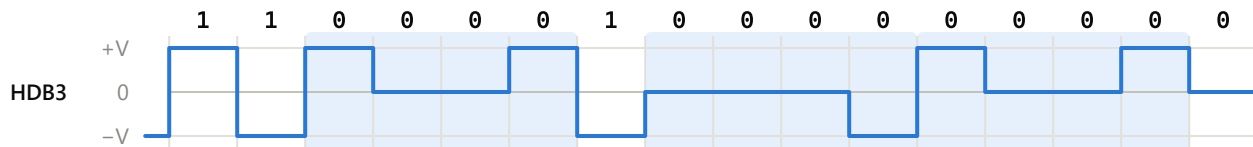


WORKED EXAMPLE

HDB3 (slide 61) L03 p61

Apply **HDB3** scrambling (on top of AMI) to the data stream **1100 0010 0000 0000** and draw the line signal. The last nonzero pulse sent before this stream was negative ($-V$). At the start of the stream the number of nonzero pulses since the last substitution is even.

- Scrambling works on top of AMI: 0 = 0 V and the 1s alternate in polarity. The last pulse before the stream was negative ($-V$), so the next 1 is +. L03 p58
- HDB3 replaces four consecutive 0s: count the nonzero pulses since the last substitution — **even** → **B00V**, **odd** → **000V** (the total becomes even either way). L03 p60
- Substitution 1 (bits 3–6): 2 nonzero pulses since the start → even → **B00V** = + 0 0 + (B alternates from the previous pulse, V repeats B). L03 p61
- Substitution 2 (bits 8–11): 1 nonzero pulse since the last substitution → odd → **000V** = 0 0 0 - (V repeats the previous pulse). L03 p61
- Substitution 3 (bits 12–15): 0 nonzero pulses since the last substitution → even → **B00V** = + 0 0 + (B alternates from the previous pulse, V repeats B). L03 p61
- The transmitted pulses are + - + 0 0 + - 0 0 0 - + 0 0 + 0 — no run of four zeros is left.



Pulse before the stream: -. Substitutions: bits 3–6 → B00V (even pulses since last substitution); bits 8–11 → 000V (odd pulses since last substitution); bits 12–15 → B00V (even pulses since last substitution). Marks: B at bit 3, V at bit 6, V at bit 11, B at bit 12, V at bit 15.

Slide's answer: + - + 0 0 + - 0 0 0 - + 0 0 + 0

Formulas

Signal (baud) rate L03 p8

$$S = c \times N \times \frac{1}{r}$$

N = data rate (bps); r = data elements per signal element; c = case factor (worst 1, best 0, average $\frac{1}{2}$)

Minimum bandwidth L03 p25; L03 p46

$$B_{\min} = S_{\text{avg}}$$

NRZ-L/NRZ-I/AMI N/2 · RZ, Manchester N · 2B1Q N/4 · MLT-3 N/3 · 8B6T 3N/4 · 4D-PAM5 N/8

Nyquist reminder L03 p10

$$N_{\max} = 2 \times B \times \log_2 L$$

the effective bandwidth of a digital signal is finite

mBnL code condition L03 p36; L03 p37

$$2^m \leq L^n$$

m data bits → n signal elements with L levels (B = 2, T = 3, Q = 4)

4B/5B rate L03 p54

$$N_{\text{line}} = N \times 5/4$$

every 4 data bits become 5 code bits

Clock drift L03 p14

extra bits/s = N × (clock error)

0.1% fast at 1 kbps → 1 extra bps

⚠ COMMON TRAPS

- **Mixing NRZ-L polarities between parts of an answer** → Pick one convention (slide figure: 0 = +V) and state it in a legend. L03 p23
- **Flipping Manchester's direction** → Slide 29 legend: 0 = high→low, 1 = low→high. Differential Manchester: a 0 changes level at the START of the bit. L03 p29
- **Inverting NRZ-I on a 0** → Only a 1 inverts, at the beginning of the bit. L03 p22
- **Using the bit rate as the bandwidth** → $B_{\min}$ follows the average signal rate: $N/2$ for NRZ, N for Manchester. L03 p25; L03 p46
- **Taking 2B1Q's r as $1/2$ (as the slide box says)** → $r = 2$ — two bits per level — so $S_{\text{avg}} = N/4$. L03 p38
- **Forgetting that 4B/5B raises the rate first** → Multiply by $5/4$ before applying the line code's bandwidth rule (1 Mbps → 1.25 Mbps → 625 kHz with NRZ-I). L03 p54
- **B8ZS: giving V the AMI polarity** → V violates AMI (same polarity as the previous pulse); B obeys AMI. L03 p59
- **HDB3: counting pulses from the start of the stream** → Count nonzero pulses since the LAST substitution: even → B00V, odd → 000V. L03 p60

💡 MNEMONICS & MEMORY HOOKS

- NRZ-I: "Invert on **one**".
- Manchester: "one goes **up**, zero goes **down**" (mid-bit). Differential Manchester: "**zero changes** at the start".
- AMI: "**zeros are zero**, ones alternate". Pseudoternary swaps the roles.
- Bandwidth ladder: 4D-PAM5 $N/8$ · 2B1Q $N/4$ · MLT-3 $N/3$ · NRZ & AMI $N/2$ · 8B6T $3N/4$ · Manchester & RZ N .
- B8ZS = "000 V B 0 V B"; HDB3 = "Even? B00V. Odd? 000V."

Blank-page recall

Close the notes. Answer each prompt from memory on paper, then check against the sections above.

1. Write $S = c \cdot N \cdot (1/r)$, say what c and r mean, and redo Examples 1 and 3.
2. List the six characteristics of a good line code and explain baseline wandering and DC components.
3. Draw 01001110 in NRZ-L and NRZ-I, and 010011 in Manchester and differential Manchester, from memory.
4. Rebuild the slide-46 summary table: scheme, average bandwidth, sync/DC behaviour.
5. Encode 0000 1111 with 4B/5B; explain why 4B/5B is paired with NRZ-I; redo Example 4.
6. Apply B8ZS to 1 00000000 and HDB3 to slide 61's stream.

⚠ SLIDE ERRORS & CONTRADICTIONS

On the exam, the answer that follows the slide's intent is the safe one — but know the correct version.

Slide	Slide says	Correct / intended
L03 p22	NRZ-L text: “+V for 1 and −V for 0” (and “NZR – Level”)	The slide-23 figure uses $0 = +V$, $1 = -V$; state your convention. “NZR” is a typo for NRZ.
L03 p5	Example mapping $1 \rightarrow +V$ then $-V$ ($1 = \text{high} \rightarrow \text{low}$)	The Manchester legend on slide 29: $0 = \text{high} \rightarrow \text{low}$, $1 = \text{low} \rightarrow \text{high}$.
L03 p38	2B1Q box: $r = 1/2$	$r = 2$ (two bits per level), giving $S_{\text{avg}} = N/4$.
L03 p46	NRZ-L, NRZ-I and biphase listed under “Unipolar”	They are polar schemes.
L03 p46	AMI marked “DC”	Slide 33: AMI has no DC component.
L03 p46	Category “Multiline” for MLT-3	Multitransition (slides 19 and 44).
L03 p53	“24 ... 25 = 32 ... 32 – 26 = 16”	$2^4 = 16$, $2^5 = 32$, $32 - 16 = 16$.
L03 p58	Receiver box: “Modified AMI encoding”	It should say decoding.
L03 p19	“2B/1Q, 8B/6T” written with a slash	The slash marks block coding (mB/nB); multilevel codes are written 2B1Q and 8B6T.
L03 p25	“ $S = c \times N \times R$ ”	$S = c \times N \times 1/r$.

Glossary

Line coding	Converting a string of bits into a sequence of signal levels. L03 p3
Data element	A data symbol: one bit or a group of bits (1, 0 or 11, 10, 01, ...) that is mapped onto signal levels. L03 p5
Signal element	The unit of the signal that carries data elements; a data symbol is coded into one or more signal elements. L03 p5
Ratio r	Number of data elements carried by each signal element. L03 p5
Signal rate	Signal elements per second (baud); $S = c \times N \times 1/r$. L03 p6; L03 p8
Case factor (c)	Worst case 1, best case 0, average $1/2$ in $S = c \times N \times 1/r$. L03 p8
Baseline wandering	Drift of the receiver's running-average power (the baseline) caused by long runs of 0s or 1s. L03 p11
DC component	Low-frequency energy from a constant voltage held for long periods; bandpass channels cannot pass it. L03 p12
Self-synchronization	Transitions in the signal that keep the receiver's clock aligned with the sender's bit intervals. L03 p13

NRZ-L	Polar NRZ where the level decides the bit (slide figure: 0 = +V, 1 = -V). L03 p22; L03 p23
NRZ-I	Polar NRZ where a 1 inverts the level and a 0 keeps it. L03 p22
RZ	Polar code with three levels; the signal returns to zero in the middle of every bit. L03 p26
Manchester	Biphase code (NRZ-L + RZ): mid-bit transition every bit; 0 = high→low, 1 = low→high. L03 p28; L03 p29
Differential Manchester	Biphase code (NRZ-I + RZ): mid-bit transition every bit; a 0 changes level at the start of the bit. L03 p28
AMI	Bipolar code: 0 = 0 V, 1s alternate between +V and -V. L03 p31
Pseudoternary	Bipolar code, the reverse of AMI: 1 = 0 V, 0s alternate. L03 p31
mBnL	Multilevel code: m data bits → n signal elements with L levels; requires $2^m \leq L^n$. L03 p36; L03 p37
2B1Q	Two bits per quaternary level (four levels), coded relative to the previous level; $S = N/4$. L03 p38
MLT-3	Multitransition code cycling through +V, 0, -V: a 0 keeps the level, a 1 moves to the next level. L03 p44
Block coding	mB/nB substitution adding redundancy for synchronization and error detection (division, substitution, combination). L03 p48
4B/5B	Block code replacing every 4 bits with a 5-bit code word; raises the rate by 5/4; used with NRZ-I. L03 p50; L03 p51
Scrambling	Replacing unfriendly bit runs on the fly with recognizable violation codes, without adding bandwidth. L03 p57
B8ZS	Bipolar with 8-zero substitution: 00000000 → 000VB0VB. L03 p59
HDB3	High-density bipolar 3-zero: 0000 → 000V (odd pulses since last substitution) or B00V (even). L03 p60
Violation (V)	A pulse that breaks the AMI rule (same polarity as the previous pulse), so the receiver recognizes a substitution. L03 p59

Digital Transmission II: PCM, Delta Modulation & Transmission Modes

Turning an analog signal into bits (sample → quantize → encode), the one-bit delta-modulation alternative, and how bits travel: parallel vs serial, asynchronous vs synchronous.

★ Nyquist: $f_s \geq 2 \cdot f_{\max}$

★ Δ , zones and codes (slides 74–76)

★ $N = n_b \times f_s$ and $B_{\min} = n_b \times f_{\max}$

★ Start/stop bits

AT A GLANCE

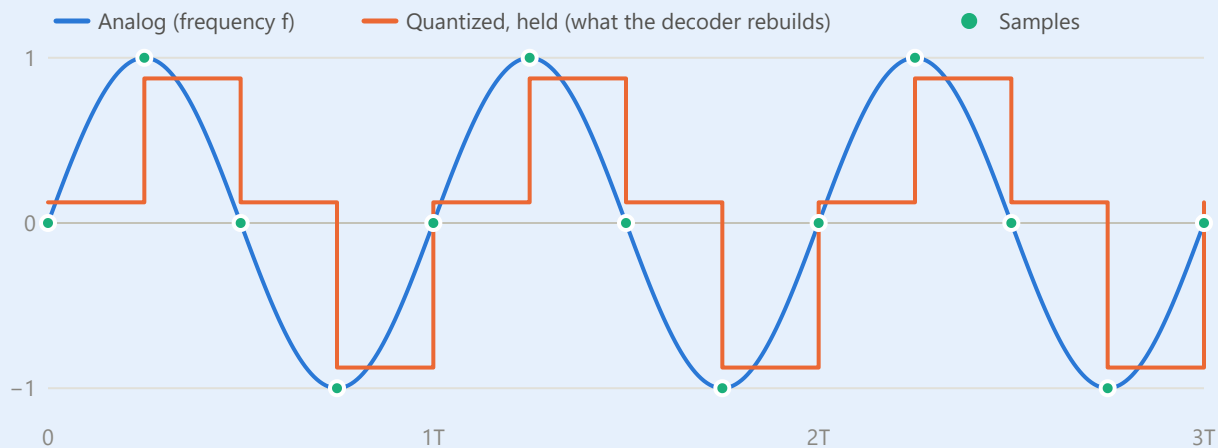
- PCM = **sampling** → **quantization** → **binary encoding**; filter first so the highest frequency is known. (Marked “Main Section 7” — expect a PCM problem.)
- Nyquist: sample at $f_s \geq 2 \times f_{\max}$ for low-pass and bandpass signals. Voice up to 4 kHz → 8000 samples/s. A bandpass signal with only its bandwidth known → the rate **cannot be determined**.
- Quantization: $\Delta = (\max - \min)/L$; zones 0 ... L–1 from the bottom; each sample becomes its zone’s **midpoint**; $n_b = \log_2 L$ bits; error $\leq \Delta/2$. More levels → less error but more bits.
- $N = n_b \times f_s$ (8-bit voice → 64 kbps). Digitizing costs bandwidth: $B_{\min} = n_b \times f_{\max}$ with an NRZ-type line code (32 kHz instead of 4 kHz).
- **Delta modulation** sends 1 bit per sample (1 = step up δ , 0 = step down δ): fine for slow changes, poor for fast ones. **DPCM** quantizes the difference with more bits.
- **Parallel**: n bits per clock tick over n wires. **Serial**: 1 bit per tick — **asynchronous** (start 0 + byte + stop 1, gaps allowed), **synchronous** (no start/stop or gaps, frames), **isochronous** (equal, fixed gaps).

WHAT THEY’LL ASK

- **Solving**: minimum sampling rate for a low-pass or bandpass signal; sampling interval.
- **Solving**: Δ , zones, code word, quantized value and error for a sample (slides 74–76 style).
- **Solving**: PCM bit rate and minimum bandwidth (the voice example), sometimes from L instead of n_b .
- **Solving**: delta-modulation bits for a sample sequence; asynchronous framing overhead.
- **MCQ / Identification**: ideal, natural and flat-top sampling; PAM; companding; hold circuit; DPCM; isochronous.
- **Essay**: explain the PCM steps and the trade-off between quantization error and bit rate; compare asynchronous and synchronous transmission.

💡 INTUITION FIRST

Snapshots and a ruler



✓ $f_s = 4f \geq 2f$ — the Nyquist condition holds.

$L = 8$ zones $\rightarrow \Delta = 0.25$ (of full scale 2), $n_b = \log_2 L = 3$ bits/sample, worst-case error $\approx \Delta/2 = 0.125$.

For telephone voice ($f_{\max} = 4$ kHz) at this sampling ratio: $f_s = 16$ ksamples/s, bit rate = $n_b \times f_s = 48$ kbps.

Sampling is a flip-book: with enough snapshots per second the motion looks smooth; with too few, the motion lies to you (like wagon wheels that seem to spin backwards in films). **Quantizing** is rounding every snapshot to the nearest tick on a ruler — more ticks mean smaller rounding errors, but every reading needs more digits (bits).

Predict: A voice signal contains frequencies up to 4 kHz. You sample it 5000 times per second. What happens?

Answer: It cannot be recovered correctly — the samples trace a false, slower wave

Nyquist requires $f_s \geq 2 \times 4$ kHz = 8000 samples/s. At 5000 the signal is undersampled, as in part (c) of slide 68 — more quantization levels cannot fix that. [L03 p66; L03 p68]

Predict: You double the quantization levels from $L = 8$ to $L = 16$ at the same sampling rate. What happens to the bit rate?

Answer: It rises by one third (3 $\rightarrow$ 4 bits per sample)

$n_b = \log_2 L$ goes from 3 to 4, so $N = n_b \times f_s$ grows by 4/3. Doubling L adds just one bit per sample. [L03 p75; L03 p79]

Pulse code modulation (PCM): three steps L03 pp62–63

PCM converts an analog signal into digital data in three steps: **sampling**, **quantization** and **binary encoding**. L03 p62

Before sampling, the signal is **filtered** to limit its maximum frequency, because $f_{\max}$ decides the sampling rate. The filter must not distort the signal — it should not remove the high-frequency components that shape it. L03 p62

The encoder chain (slide 63): analog signal $\rightarrow$ **sampling** $\rightarrow$ PAM signal $\rightarrow$ **quantizing** $\rightarrow$ quantized signal $\rightarrow$ **encoding** $\rightarrow$ digital data — the figure in “Intuition first” above. L03 p63

★ HIGH YIELD

The instructor marked PCM “Main Section 7” and its worked examples as sub-sections — expect a PCM computation.

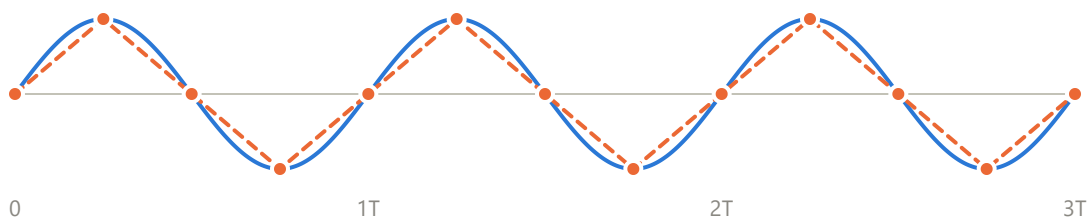
Sampling and the Nyquist theorem L03 pp64–68

The analog signal is sampled every T_s seconds (the sampling interval); $f_s = 1/T_s$ is the sampling rate. The process is called **pulse amplitude modulation (PAM)**, and its output still has analog (non-integer) values. L03 p64

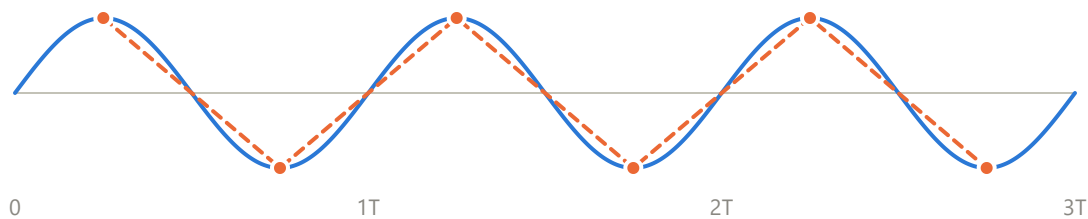
Method	Each sample is ...
Ideal	an impulse at the sampling instant
Natural	a short pulse whose top follows the signal
Flat-top	a short pulse held at one amplitude (sample and hold)

Nyquist theorem: the sampling rate must be at least **2 times the highest frequency** in the signal: $f_s \geq 2 \cdot f_{\max}$. L03 p66 The slides give the Nyquist rate as $2 \times f_{\max}$ for both **low-pass** and **bandpass** signals — so you must know $f_{\max}$. L03 p67

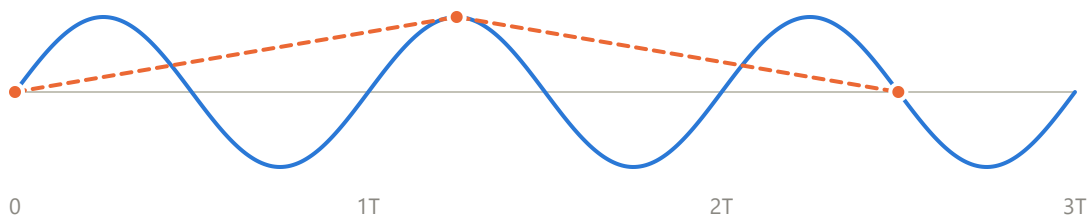
Oversampling: $f_s = 4f$ — the same shape, with redundant samples



Nyquist rate: $f_s = 2f$ — just enough to rebuild the wave



Undersampling: f_s below $2f$ — the samples trace a slower, false wave (the slide labels this $f_s = f$)



Slide 68: sampling a sine wave at $4f$, at the Nyquist rate $2f$, and below it.

Sampling at the Nyquist rate gives a good approximation; oversampling gives the same approximation but is redundant; undersampling does not produce a signal that looks like the original. L03 p68

WORKED EXAMPLE

Telephone voice L03 p69

A low-pass analog signal contains frequencies from 0 to **4 kHz** (bandwidth 4 kHz). What is the minimum sampling rate, and what is the sampling interval at that rate?

1. A low-pass signal runs from 0 to $f_{\max}$, so $f_{\max} = 4 \text{ kHz}$. L03 p70
2. Nyquist: $f_s \geq 2 \times f_{\max} = 2 \times 4 \text{ kHz} = \mathbf{8 \text{ ksamples/s}}$. L03 p66
3. $T_s = 1 / f_s = \mathbf{125 \text{ }\mu\text{s}}$. L03 p64

Minimum $f_s = 8 \text{ ksps}$ Sampling interval $T_s = 125 \text{ }\mu\text{s}$

Slide's answer: 8000 samples per second

WORKED EXAMPLE

Low-pass signal with a 200 kHz bandwidth L03 p70

A low-pass analog signal contains frequencies from 0 to **200 kHz** (bandwidth 200 kHz). What is the minimum sampling rate, and what is the sampling interval at that rate?

1. A low-pass signal runs from 0 to $f_{\max}$, so $f_{\max} = 200 \text{ kHz}$. L03 p70
2. Nyquist: $f_s \geq 2 \times f_{\max} = 2 \times 200 \text{ kHz} = \mathbf{400 \text{ ksamples/s}}$. L03 p66
3. $T_s = 1 / f_s = \mathbf{2.5 \text{ }\mu\text{s}}$. L03 p64

Minimum $f_s = 400 \text{ ksps}$ Sampling interval $T_s = 2.5 \text{ }\mu\text{s}$

Slide's answer: 400,000 samples per second

WORKED EXAMPLE

Bandpass signal with a 200 kHz bandwidth L03 p71

A complex **bandpass** signal has a bandwidth of **200 kHz**. What is the minimum sampling rate for this signal?

1. The Nyquist rate is $2 \times f_{\max}$. L03 p67
2. For a bandpass signal we only know the width of the band, not where it starts or ends — so $f_{\max}$ is unknown.
3. **The minimum sampling rate cannot be determined.** (Slide Example 8.) L03 p71

Answer = **It cannot be determined**

Slide's answer: cannot be determined — we do not know where the band starts or ends

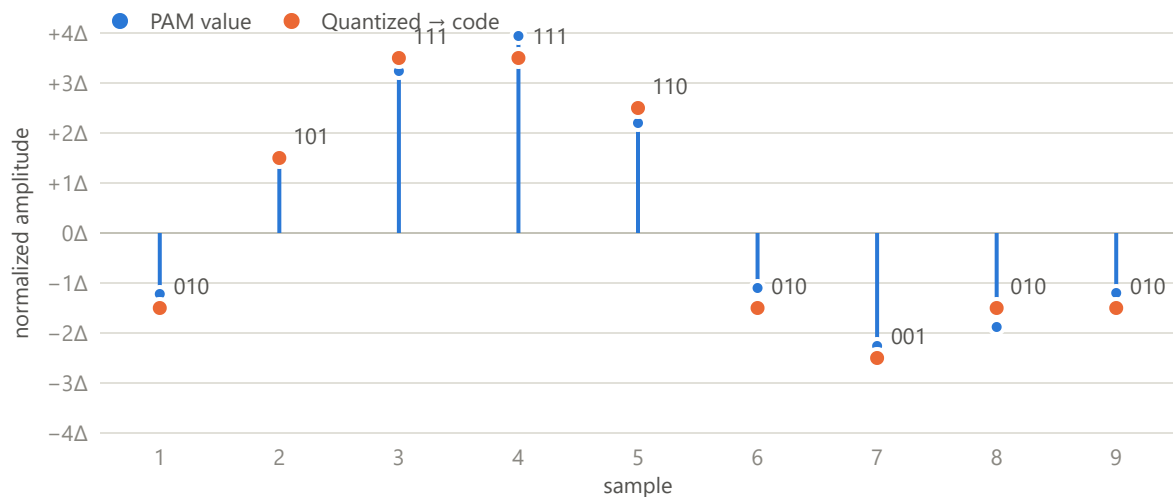
Quantization and encoding L03 pp72–78

Sampling gives pulses whose amplitudes can be any value between a minimum and a maximum. Quantization maps them onto a finite set: divide the range into **L zones** of height $\Delta = (\text{max} - \text{min})/L$. L03 p72 Each zone's **midpoint** is the value that every sample in that zone becomes; zones are numbered 0 to $L - 1$. L03 p73

Slide example: -20 V to $+20 \text{ V}$ with $L = 8 \rightarrow \Delta = (20 - (-20))/8 = \mathbf{5 \text{ V}}$. Zones: -20 to -15 , -15 to -10 , ..., $+15$ to $+20$; midpoints -17.5 , -12.5 , -7.5 , -2.5 , 2.5 , 7.5 , 12.5 , 17.5 . L03 p74

Encoding: $n_b = \log_2 L$ bits per sample — here 3. The codes run 000 (-20 to -15 V), 001 (-15 to -10 V), ..., 111 ($+15$ to $+20 \text{ V}$).

L03 p75



Slide 76: nine samples, their zone midpoints and 3-bit codes.

Original amplitude (V)	-6.1	7.5	16.2	19.7	11	-5.5	-11.3	-9.4	-6
Normalized PAM value	-1.22	1.5	3.24	3.94	2.2	-1.1	-2.26	-1.88	-1.2
Normalized quantized value	-1.5	1.5	3.5	3.5	2.5	-1.5	-2.5	-1.5	-1.5
Normalized error	-0.28 Δ	0	+0.26	-0.44	+0.3	-0.4	-0.24	+0.38	-0.3
Quantization code	2	5	7	7	6	2	1	2	2
Encoded word	010	101	111	111	110	010	001	010	010

⚠ SLIDE SAYS / CORRECT

Slide 76 first normalized error -0.38

Correct $-1.50 - (-1.22) = -0.28$. Every other cell checks out.

Quantization error

- The coded value (midpoint) is only an approximation: the difference is the **quantization error**, at most $\Delta/2$ in size. L03 p77
- More zones $\rightarrow$ smaller $\Delta \rightarrow$ smaller error, **but** more bits per sample $\rightarrow$ higher bit rate. L03 p77
- Because the error range $\Delta/2$ is fixed, **low-amplitude signals suffer more** (worse SNQR). Fixes: **non-linear quantization** (smaller Δ at low amplitudes, logarithmic zones) and **companding** (compress at the sender, expand at the receiver). L03 p78
- Beyond slides Textbook rule of thumb: $\text{SNQR} \approx 6.02 n_b + 1.76 \text{ dB}$ — each extra bit adds about 6 dB.

WORKED EXAMPLE

Quantize –6.1 V (the first sample of slide 76) L03 p74; L03 p76

A signal swings between **–20 V** and **20 V** and is quantized into **L = 8** zones (zones numbered 0 to 7 from the bottom, each sample replaced by its zone's midpoint, as in the slides). A sample has the value **–6.1 V**. Find Δ , its code word, its quantized value, and the quantization error (quantized – actual).

1. $\Delta = (\text{max} - \text{min}) / L = (20 - (-20)) / 8 = \mathbf{5\text{ V}}$. L03 p72
2. $n_b = \log_2 8 = 3$ bits per sample; zone k is sent as k in binary. L03 p75
3. **–6.1 V** falls in zone 2: **–10 V** to **–5 V** → code **010**. L03 p74
4. The sample is replaced by the zone midpoint: **–7.5 V**. L03 p73
5. Error = quantized – actual = $-7.5 - (-6.1) = \mathbf{-1.4\text{ V}}$ ($|\text{error}| \leq \Delta/2 = 2.5\text{ V}$). L03 p76

$\Delta = \mathbf{5\text{ V}}$ Code word = **010** Quantized value = **–7.5 V** Quantization error = **–1.4 V**

Slide's answer: zone 2 → code 010, quantized **–7.5 V** (normalized **–1.50**), error **–1.4 V** (normalized **–0.28**)

PCM bit rate, bandwidth and decoding L03 pp79–83

Bit rate = $n_b \times f_s$. The bandwidth needed depends on the line code (see L03a). A digitized signal always needs **more bandwidth** than the original analog signal — the price of robustness and the other features of digital transmission. L03 p79

- **Example 9:** voice (0–4000 Hz), 8 bits per sample → $f_s = 8000$ samples/s → $N = 8000 \times 8 = \mathbf{64\text{ kbps}}$. L03 p80
- **Example 10:** sending the 4 kHz analog signal needs 4 kHz; digitized with 8 bits per sample it needs at least $8 \times 4\text{ kHz} = \mathbf{32\text{ kHz}}$ ($B_{\min} = N/2$ for an NRZ-type code). L03 p83

PCM decoder: a **hold circuit** keeps each pulse's amplitude until the next pulse arrives (a staircase), then a **low-pass filter** with cutoff equal to the highest frequency of the pre-sampled signal smooths it. The higher L , the less distorted the recovered signal.

L03 p81 L03 p82

WORKED EXAMPLE

Digitized voice: bit rate and bandwidth L03 p80; L03 p83

A low-pass analog signal with frequencies up to **4 kHz** is digitized with PCM using **8 bits per sample**, sampled at the Nyquist rate. Find the sampling rate, the bit rate, and the minimum bandwidth of the digitized signal (NRZ-type line code).

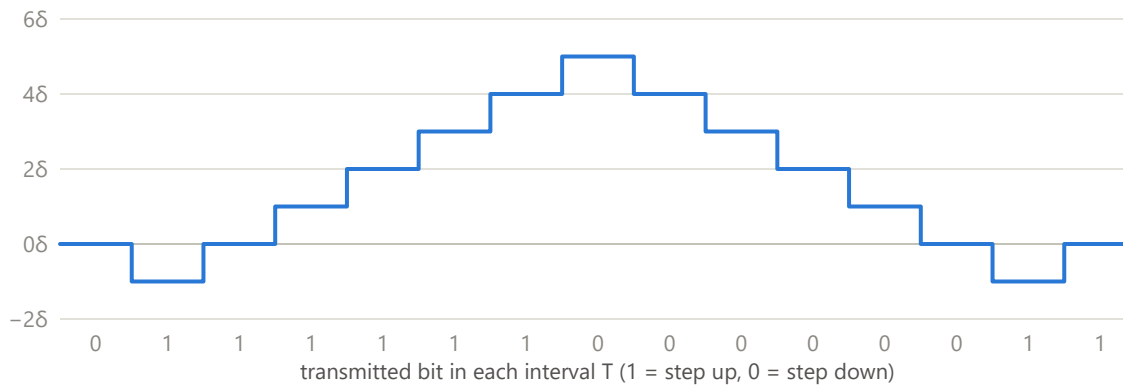
1. $f_{\max} = 4\text{ kHz} \rightarrow$ Nyquist sampling rate $f_s = 2 \times f_{\max} = \mathbf{8\text{ ksamples/s}}$. L03 p69
2. Bit rate $N = n_b \times f_s = 8 \times 8\text{ ksamples/s} = \mathbf{64\text{ kbps}}$. L03 p79
3. Minimum bandwidth (NRZ-type line code, $c = 1/2$, $r = 1$): $B_{\min} = N/2 = n_b \times f_{\max} = \mathbf{32\text{ kHz}}$ — versus only 4 kHz for the analog signal. L03 p83

Sampling rate $f_s = \mathbf{8\text{ ksps}}$ Bit rate $N = \mathbf{64\text{ kbps}}$ Minimum bandwidth = **32 kHz**

Slide's answer: 8000 samples/s; 64 kbps (slide 80); minimum bandwidth 32 kHz (slide 83)

Delta modulation (DM) and DPCM L03 pp84–88

Delta modulation sends **only the change** between samples: if the next sample is higher than the current staircase, send **1** (step up by δ); if lower — or equal — send **0** (step down by δ). Slide 84 words it as comparing each pulse with the previous pulse; the slide-85 figure, like the textbook, compares the sample with the staircase — the two agree while the staircase keeps up with the signal. It works well for small changes between samples, but large changes cause large errors. L03 p84



Slide 85: the generated bits 0 1 1 1 1 1 1 0 0 0 0 0 1 1 — one step down, six up, six down, two up.

- **Modulator:** comparator + staircase maker + delay unit (the delayed staircase is compared with the next sample). L03 p86
- **Demodulator:** staircase maker + delay unit + low-pass filter. L03 p87
- **DPCM:** instead of one bit, quantize the difference with several bits — more bits, more levels, higher accuracy. L03 p88

WORKED EXAMPLE

Delta-modulation bits L03 p85

A delta modulator starts with its staircase at **2** and uses step size $\delta = 1$. The samples are: **1.6, 1.8, 2.5, 3.6, 4.5, 5.5, 6.3, 6.4, 5.6, 4.4, 3.5, 2.6, 1.8, 1.4, 2.3**. Give the transmitted bits and the final staircase value.

1. Rule: if the sample is higher than the current staircase, send 1 and step up by δ ; otherwise send 0 and step down by δ .

L03 p84

2. Sample 1.6 $\leq$ staircase 2 $\rightarrow$ **0**, staircase $-1 = 1$.
3. Sample 1.8 $>$ staircase 1 $\rightarrow$ **1**, staircase $+1 = 2$.
4. Sample 2.5 $>$ staircase 2 $\rightarrow$ **1**, staircase $+1 = 3$.
5. Sample 3.6 $>$ staircase 3 $\rightarrow$ **1**, staircase $+1 = 4$.
6. Sample 4.5 $>$ staircase 4 $\rightarrow$ **1**, staircase $+1 = 5$.
7. Sample 5.5 $>$ staircase 5 $\rightarrow$ **1**, staircase $+1 = 6$.
8. Sample 6.3 $>$ staircase 6 $\rightarrow$ **1**, staircase $+1 = 7$.
9. Sample 6.4 $\leq$ staircase 7 $\rightarrow$ **0**, staircase $-1 = 6$.
10. Sample 5.6 $\leq$ staircase 6 $\rightarrow$ **0**, staircase $-1 = 5$.
11. Sample 4.4 $\leq$ staircase 5 $\rightarrow$ **0**, staircase $-1 = 4$.
12. Sample 3.5 $\leq$ staircase 4 $\rightarrow$ **0**, staircase $-1 = 3$.
13. Sample 2.6 $\leq$ staircase 3 $\rightarrow$ **0**, staircase $-1 = 2$.
14. Sample 1.8 $\leq$ staircase 2 $\rightarrow$ **0**, staircase $-1 = 1$.
15. Sample 1.4 $>$ staircase 1 $\rightarrow$ **1**, staircase $+1 = 2$.
16. Sample 2.3 $>$ staircase 2 $\rightarrow$ **1**, staircase $+1 = 3$.
17. Bits: **011111100000011**; final staircase value **3**.

L03 p85

Transmitted bits = **011111100000011** Final staircase value = **3**

Slide's answer: 011111100000011 — the samples were chosen to reproduce the slide's generated bits

Transmission modes: parallel and serial L03 pp89–96

Binary data crosses a link in **parallel** or **serial** mode. L03 p89

Mode	How	Notes
Parallel	multiple bits with each clock tick	8 bits sent together need 8 lines L03 p90
Serial	1 bit with each clock tick	one line; parallel/serial converters at each end L03 p91
↳ Asynchronous	1 start bit (0) at the beginning and 1 or more stop bits (1) at the end of every byte	gaps between bytes allowed; “asynchronous at the byte level”, but bits inside a byte keep equal durations L03 p92
↳ Synchronous	bits one after another, no start/stop bits, no gaps	the receiver groups the bits; bytes are grouped into frames marked by start and end bytes L03 p94
↳ Isochronous	no uneven gaps between frames	bits sent at fixed, equal gaps L03 p96

Asynchronous 

Synchronous 

 start bit (0)  data byte  stop bit (1)

Slide 93: start bit first on the line, then the data, then the stop bit; gaps between bytes. Synchronous frames pack bytes back to back.

WORKED EXAMPLE

Asynchronous framing overhead L03 p92

Characters of **8 data bits** are sent asynchronously with one start bit and **1 stop bit** over a **2.4 kbps** line, with no gaps. Find the bits per character frame, the efficiency, the characters per second, and the time to send 480 characters.

- Each character is framed by 1 start bit (0) and 1 stop bit (1): $1 + 8 + 1 = \mathbf{10 \text{ bits}}$. L03 p92
- Efficiency = data bits / frame bits = $8/10 = \mathbf{80\%}$ (the rest is framing overhead).
- Characters per second = $2.4 \text{ kbps} / 10 = \mathbf{240}$.
- Time = $480 / 240 = \mathbf{2 \text{ s}}$.

Bits per frame = 10 Efficiency (ratio or %) = 0.8 Characters per second = 240 Time for 480 characters = 2 s

Formulas

Sampling rate L03 p64

$$f_s = 1 / T_s$$

T_s = sampling interval (s); f_s in samples/s

Nyquist sampling theorem L03 p66; L03 p67

$$f_s \geq 2 \times f_{\max}$$

$f_{\max}$ = highest frequency in the signal — for low-pass and bandpass signals alike

Zone height (quantization step) L03 p72

$$\Delta = (\max - \min) / L$$

L = number of zones (levels)

Bits per sample L03 p75

$$n_b = \log_2 L$$

L is a power of 2 in the slides; otherwise round up (textbook)

PCM bit rate L03 p79

$$N = n_b \times f_s$$

bits/sample $\times$ samples/s

Minimum PCM bandwidth L03 p83

$$B_{\min} = n_b \times f_{\max}$$

NRZ-type line code ($c = 1/2$, $r = 1$): $B_{\min} = N/2$
Slide example 10: $8 \times 4 \text{ kHz} = 32 \text{ kHz}$.

Quantization error bound L03 p76; L03 p78

$$|\text{error}| \leq \Delta / 2$$

error = quantized (midpoint) – actual

SNQR (textbook) Forouzan **Beyond slides**

$$\text{SNQR}_{\text{dB}} \approx 6.02 n_b + 1.76$$

each extra bit per sample adds about 6 dB

Asynchronous efficiency L03 p92; derived

$$\text{efficiency} = \text{data bits} / (\text{start} + \text{data} + \text{stop bits})$$

8 data bits + 1 start + 1 stop $\rightarrow 8/10 = 80\%$

Worked out from the slide-92 frame; the slides give no formula.

⚠ COMMON TRAPS

- **Using the bandwidth of a bandpass signal as $f_{\max}$** $\rightarrow$ Nyquist needs the highest frequency; with only the bandwidth known, the rate cannot be determined. L03 p71
- **Thinking that doubling L doubles the bit rate** $\rightarrow$ Doubling L adds one bit per sample ($n_b = \log_2 L$). L03 p75
- **Numbering zones from the top, or starting at 1** $\rightarrow$ Zones are 0 ... L – 1 from the bottom; 000 is the lowest zone (–20 to –15 V on slide 75). L03 p75
- **Quantizing to a zone edge** $\rightarrow$ Each sample becomes its zone's **midpoint**. L03 p73
- **Assuming the digitized signal needs only $f_{\max}$ of bandwidth** $\rightarrow$ It needs $n_b \times f_{\max}$ — 32 kHz for 8-bit voice, not 4 kHz. L03 p83
- **Start bit = 1, stop bit = 0** $\rightarrow$ The start bit is 0 and the stop bits are 1. L03 p92
- **“Asynchronous” means the bits are not synchronized** $\rightarrow$ It is asynchronous at the byte level (gaps between bytes); bits inside a byte still have equal durations. L03 p92

💡 MNEMONICS & MEMORY HOOKS

- PCM = **S**ample, **Q**uantize, **E**ncode — “Some Quiet Engineers”.
- Nyquist doubles: highest frequency $\times 2$ = the fewest samples per second.
- Δ = range $\div$ levels $\cdot$ bits = $\log_2$ levels $\cdot$ bit rate = bits $\times$ samples/s.
- Asynchronous: “start low, stop high” — a 0 opens every byte, a 1 closes it.
- Ideal / natural / flat-top = impulse / follows the curve / holds one value.

Blank-page recall

Close the notes. Answer each prompt from memory on paper, then check against the sections above.

1. List the three PCM steps and what the signal looks like after each (PAM → quantized → bits).
2. State the Nyquist theorem. Why can't slide 71's bandpass signal be given a sampling rate?
3. Rebuild slides 74–75 from memory: zones, midpoints and codes for $-20 \dots +20$ V with $L = 8$.
4. Quantize -6.1 V from slide 76: zone, code, quantized value, error.
5. Compute the bit rate and minimum bandwidth for 8-bit voice (0–4 kHz).
6. Explain delta modulation and when it fails. What does DPCM change?
7. Compare asynchronous, synchronous and isochronous serial transmission.

⚠ SLIDE ERRORS & CONTRADICTIONS

On the exam, the answer that follows the slide's intent is the safe one — but know the correct version.

Slide	Slide says	Correct / intended
L03 p62	"Filtering should ensure that we do not distort the signal, ie remove high frequency components that affect the signal shape"	Read it as: the filter must not distort the signal, so it must not remove the high-frequency components that shape it — it only limits $f_{\max}$ for sampling.
L03 p76	First normalized error -0.38	$-1.50 - (-1.22) = -0.28$. All other cells are correct.
L03 p77	Quantization error = "the difference between actual and coded value"	The slide-76 table uses quantized – actual ($3.24 \rightarrow 3.50$ gives $+0.26$). State your sign; either way $ \text{error} \leq \Delta/2$.
L03 p68	Undersampling is labelled $f_s = f$, but its samples sit at a peak, a zero and a trough	Sampling at exactly $f_s = f$ would give the same value every time; the drawing matches a rate slightly below f . The lesson holds: below $2f$ the samples trace a false, slower wave.

Glossary

PCM	Pulse code modulation: digitizing an analog signal by sampling, quantization and binary encoding. L03 p62
Sampling interval (T_s)	Time between samples; $f_s = 1/T_s$. L03 p64
Sampling rate (f_s)	Samples taken per second. L03 p64
Ideal sampling	Each sample is an impulse at the sampling instant. L03 p64
Natural sampling	Each sample is a short pulse whose top follows the signal. L03 p64
Flat-top sampling	Each sample is a short pulse held at a single amplitude (sample and hold). L03 p64
PAM	Pulse amplitude modulation — the sampled signal, with analog (non-integer) pulse heights. L03 p64
Nyquist theorem	The sampling rate must be at least twice the highest frequency in the signal. L03 p66
Quantization	Mapping the infinite range of sample values onto L zones; each sample becomes its zone midpoint. L03 p72; L03 p73

Zone height (Δ)	$\Delta = (\text{max} - \text{min})/L$ — the size of each quantization zone. L03 p72
Quantization error	The difference between the actual sample and its coded (midpoint) value; at most $\Delta/2$. L03 p77
SNQR	Signal-to-quantization-noise ratio; worse for low-amplitude signals because the error range is fixed. L03 p78
Non-linear quantization	Zones that follow a logarithmic curve: smaller Δ at low amplitudes, larger at high amplitudes. L03 p78
Companding	Compressing sample values at the sender (logarithmic zones) and expanding them at the receiver. L03 p78
Hold circuit	PCM decoder stage that holds each pulse amplitude until the next pulse arrives. L03 p81
Delta modulation	Sends one bit per sample: 1 if the signal is above the staircase (step up δ), 0 if not (step down δ). L03 p84
DPCM	Quantizes the difference between samples with several bits for higher accuracy. L03 p88
Parallel transmission	Several bits sent with each clock tick, one per line. L03 p89; L03 p90
Serial transmission	One bit sent with each clock tick over a single line. L03 p89; L03 p91
Asynchronous transmission	Each byte framed by a start bit (0) and one or more stop bits (1); gaps allowed between bytes. L03 p92
Synchronous transmission	Bits sent back to back without start/stop bits or gaps; bytes grouped into frames. L03 p94
Isochronous transmission	No uneven gaps between frames — bits sent at fixed, equal intervals. L03 p96

Digital-to-Analog: ASK, FSK, PSK & QAM

How a carrier wave is changed to carry bits — and how bit rate, baud rate, levels and bandwidth trade off.

★ $S = N/r$ and $r = \log_2 L$

★ Carrier + bit rate in a band (ASK/FSK)

★ QPSK / MFSK bandwidth

★ Reading constellations

AT A GLANCE

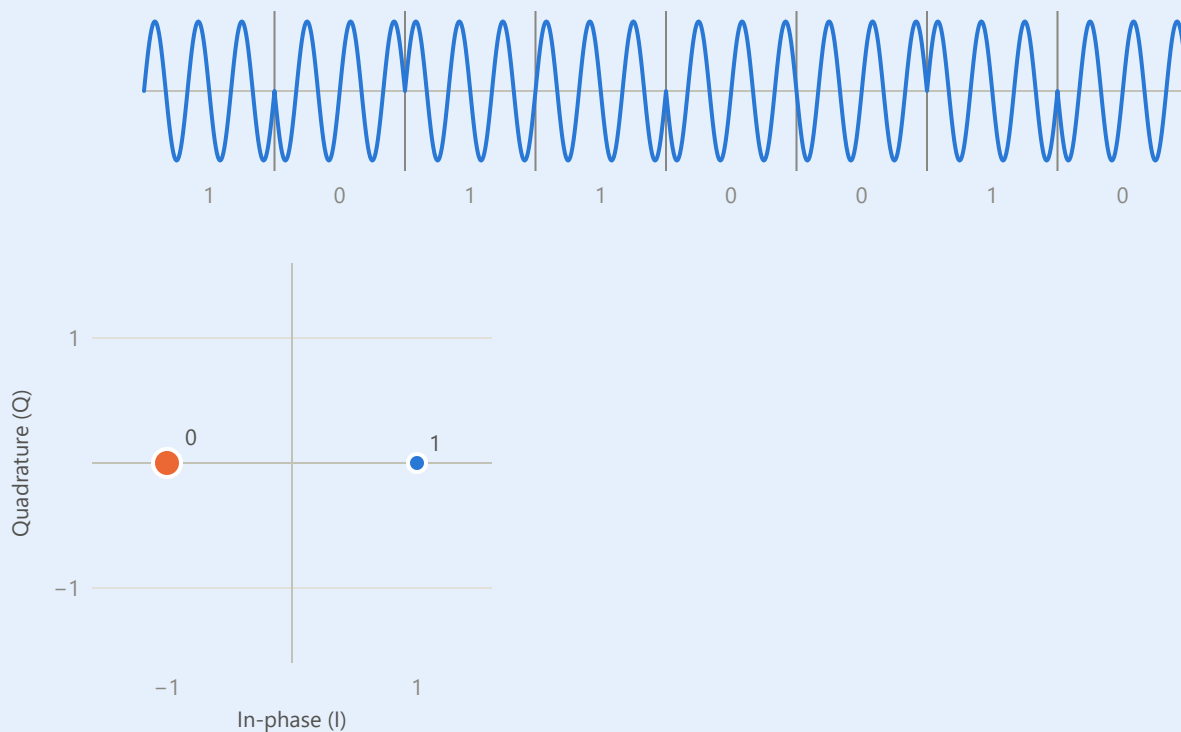
- A **carrier** of frequency f_c is modified to carry bits: change its **amplitude** (ASK), **frequency** (FSK), **phase** (PSK), or amplitude and phase together (**QAM**).
- N = bit rate, S = baud rate (signal elements/s), r = bits per element: $S = N/r$ and $r = \log_2 L$. In analog transmission, $S \leq N$.
- Bandwidth follows the **baud rate**: ASK/PSK $B = (1 + d)S$; FSK adds the spacing, $B = (1 + d)S + 2\Delta f$; MFSK $B = (1 + d)S + (L - 1)2\Delta f$ ($= L \times S$ at minimum spacing).
- Put the carrier in the **middle** of the band; for **full duplex**, split the band in two (200–300 kHz → carriers at 225 and 275 kHz).
- **PSK is more robust than ASK** (noise mostly changes amplitude). **QPSK** sends 2 bits per element on two carriers 90° apart (I and Q).
- A **constellation diagram** plots each signal element: distance from the origin = amplitude, angle = phase.

WHAT THEY'LL ASK

- **Solving**: “ r bits per element at S baud — find N ” or “ N and S given — find r and L ”.
- **Solving**: “Band from f_1 to f_2 , ASK/FSK with $d = \dots$ — find the carrier and the bit rate” (sometimes full duplex).
- **Solving**: QPSK / 16-QAM baud rate and bandwidth; MFSK levels, baud rate and bandwidth.
- **MCQ / Identification**: which scheme varies what; OOK; coherent vs non-coherent FSK; I and Q carriers; why PSK beats ASK on noise.
- **Essay**: compare ASK, FSK and PSK (what changes, bandwidth, noise robustness); explain the bit-rate/ baud-rate trade-off.

💡 INTUITION FIRST

Three knobs on one carrier



$r = 1$ bit per signal element $\rightarrow$ 8 signal elements for 8 bits. Highlighted: the last symbol, 0.

Think of the carrier as a steady musical note. To send bits you can change how **loud** it is (ASK), its **pitch** (FSK), or **when its peaks happen** — its phase (PSK). QAM changes loudness *and* phase at once, so each note can carry more bits.

Predict: Noise mostly adds random changes to a signal's **amplitude**. Which scheme is *more* robust to noise?

Answer: PSK

PSK keeps the amplitude constant and puts the data in the phase, so amplitude noise does much less damage. The slide: "PSK is much more robust than ASK as it is not that vulnerable to noise, which changes amplitude." [L04 p21]

Predict: Switch from BPSK to QPSK and keep the **baud rate** the same. What happens to the bit rate?

Answer: It doubles

QPSK carries $r = 2$ bits per signal element ($L = 4$) while BPSK carries 1. With the same S , $N = r \times S$ doubles — and the bandwidth, which follows S , stays the same. [L04 p24; L04 p26]

Digital-to-analog conversion L04 pp2-4

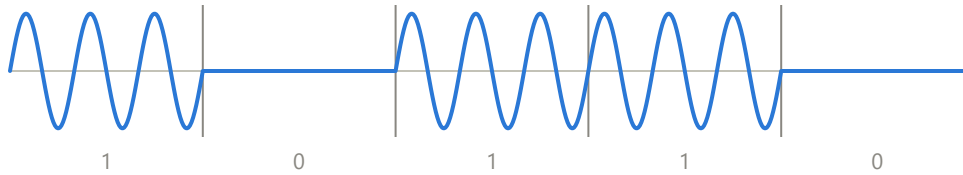
Digital data sometimes has to travel as an **analog** signal — for example over a **bandpass** channel that does not start at 0 Hz L02 p18. A **carrier signal** of frequency f_c transports the data: the sender's **modulator** changes the carrier to encode the bits and the receiver's **demodulator** recovers them. L04 p2 L04 p3

A sine wave has three characteristics (amplitude, frequency, phase — L02), so there are three basic schemes, plus QAM, which combines two: L04 p4

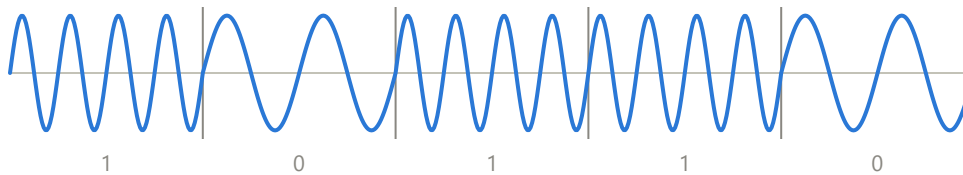
Scheme	What changes	What stays fixed
ASK — amplitude shift keying	peak amplitude	frequency, phase

Scheme	What changes	What stays fixed
FSK — frequency shift keying	frequency	amplitude, phase
PSK — phase shift keying	phase	amplitude, frequency
QAM — quadrature amplitude modulation	amplitude <i>and</i> phase (ASK + PSK)	frequency

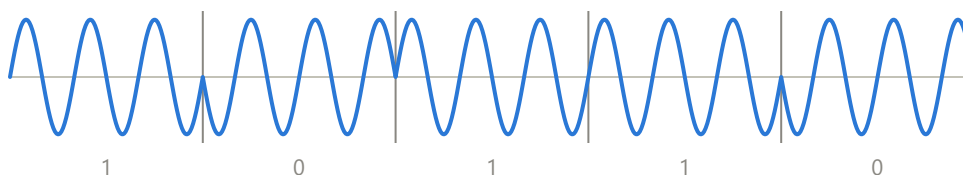
ASK (OOK)



Binary FSK



BPSK



The bits 1 0 1 1 0 on ASK (OOK), binary FSK and BPSK — compare slides 10, 14 and 22.

Bit rate vs baud rate L04 pp5–7

Bit rate N is the number of bits per second (bps). **Baud rate S** is the number of **signal elements** per second (baud) — also called the signal or modulation rate. If each element carries **r** bits: $S = N \times 1/r$. L04 p5

An element that can take **L** different forms carries $r = \log_2 L$ bits, so $L = 2^r$. In analog transmission of digital data the baud rate is **less than or equal to** the bit rate. L04 p7

WHY IT MATTERS

Bandwidth depends on the **baud rate**. Packing more bits into each element (larger **r**) raises the bit rate without widening the band for ASK, PSK and QAM (MFSK is the exception: its band grows with **L**, $B = L \times S$) — the price is more levels that the receiver must tell apart.

WORKED EXAMPLE

Bit rate from baud rate L04 p6

An analog signal carries **4 bits** in each signal element. If **1,000** signal elements are sent per second, find the bit rate.

1. Given: $r = 4$ bits per signal element, $S = 1$ kbaud.

2. $S = N \times 1/r$, so $N = S \times r$. L04 p5

3. $N = 1,000 \times 4 = 4$ **kbps**.

Bit rate **N = 4 kbps**

Slide's answer: 4000 bps

✎ WORKED EXAMPLE

Bits per element and number of levels L04 p7

An analog signal has a bit rate of **8 kbps** and a baud rate of **1 kbaud**. How many data elements are carried by each signal element? How many different signal elements (levels) do we need?

1. $S = N \times 1/r$, so $r = N / S = 8,000 / 1,000 = \mathbf{8 \text{ bits/ baud}}$. L04 p7

2. $r = \log_2 L$, so $L = 2^r = 2^8 = \mathbf{256}$.

r (bits per signal element) = 8 L (signal levels) = 256

Slide's answer: $r = 8 \text{ bits/ baud}$, $L = 256$

Amplitude shift keying (ASK) L04 pp8–13

ASK changes the **amplitude** of the carrier to reflect the data. In **binary ASK (BASK)**, a 1 leaves the carrier on and a 0 makes it zero — this is **on-off keying (OOK)**. L04 p8 L04 p10

- **Bandwidth:** $B = (1 + d) \times S$, where d (between 0 and 1) comes from modulation and filtering. BASK has $r = 1$, so $S = N$. L04 p9
- **Implementation:** multiply a **unipolar NRZ** signal by the carrier from an oscillator. L04 p11
- **Carrier choice:** put f_c in the middle of the available band. L04 p12
- **Full duplex:** both directions need a carrier, so split the band in two — each direction gets half the bandwidth. L04 p13

✎ WORKED EXAMPLE

ASK in a 200–300 kHz band L04 p12

We have an available bandwidth of **100 kHz which spans from 200 kHz to 300 kHz**. What are the carrier frequency and the bit rate if we modulate our data using **ASK** with $d = 1$?

1. The carrier goes in the middle of the band: $f_c = (200 \text{ kHz} + 300 \text{ kHz}) / 2 = \mathbf{250 \text{ kHz}}$. L04 p12

2. ASK: $B = (1 + d) \times S$. L04 p9

3. $100 \text{ kHz} = (1 + 1) \times S \Rightarrow S = 50 \text{ kbaud}$.

4. Binary ASK carries $r = 1$ bit per element, so $N = S = \mathbf{50 \text{ kbps}}$.

Carrier $f_c = 250 \text{ kHz}$ Bit rate $N = 50 \text{ kbps}$

Slide's answer: $f_c = 250 \text{ kHz}$, $N = 50 \text{ kbps}$

✎ WORKED EXAMPLE

Full-duplex ASK in the same band L04 p13

We have an available bandwidth of **100 kHz which spans from 200 kHz to 300 kHz** for **full-duplex** communication using ASK with $d = 1$. Find the two carrier frequencies and the bit rate in each direction.

1. Full duplex splits the band in two: each direction gets $100 \text{ kHz} / 2 = 50 \text{ kHz}$. L04 p13

2. Each carrier sits in the middle of its half: $f_{c1} = \mathbf{225 \text{ kHz}}$, $f_{c2} = \mathbf{275 \text{ kHz}}$.

3. ASK: $B = (1 + d) \times S$, and with $r = 1$, $S = N$. L04 p9

4. $50 \text{ kHz} = (1 + 1) \times S \Rightarrow S = 25 \text{ kbaud} \Rightarrow N = \mathbf{25 \text{ kbps}}$ each way.

Carrier 1 (lower) = 225 kHz Carrier 2 (upper) = 275 kHz Bit rate per direction = 25 kbps

Slide's answer: carriers at 225 kHz and 275 kHz; 25 kbps each way

Frequency shift keying (FSK) L04 pp14–20

FSK changes the **frequency** of the carrier. Binary FSK uses two frequencies around f_c : **bit 1** $\rightarrow f_1 = f_c + \Delta f$ and **bit 0** $\rightarrow f_2 = f_c - \Delta f$. L04 p14

- **Bandwidth:** $B = (1 + d) \times S + 2\Delta f$, where $2\Delta f$ is the gap between f_1 and f_2 . Binary FSK has $r = 1$, so $S = N$. L04 p15
- **Non-coherent FSK** switches frequency without regard to the current phase; **coherent FSK** switches only at the same phase, so the wave stays continuous. L04 p17
- **Implementation:** a voltage-controlled oscillator (VCO) changes its frequency with the NRZ input. L04 p18
- **Multilevel FSK (MFSK):** send r bits per element with $L = 2^r$ frequencies. $B = (1 + d) \times S + (L - 1) \times 2\Delta f$; with $d = 0$ and the minimum spacing $2\Delta f = S$, **$B = L \times S$** . L04 p18 L04 p19

△ SLIDE SAYS / CORRECT

Slide 18 $B = (1+d) \times S + (L-1)/2\Delta f = L \times S$

Correct $B = (1+d) \times S + (L-1) \times 2\Delta f$, which equals $L \times S$ only when $d = 0$ and $2\Delta f = S$ (the slide example's case).

✏ WORKED EXAMPLE

FSK in a 200–300 kHz band L04 p16

We have an available bandwidth of **100 kHz which spans from 200 kHz to 300 kHz**. What should be the carrier frequency and the bit rate if we modulate our data with **FSK**, $d = 1$, and $2\Delta f = 50$ kHz? Also give the frequencies used for bit 1 and bit 0.

1. Middle of the band: $f_c = (200 \text{ kHz} + 300 \text{ kHz}) / 2 = \mathbf{250 \text{ kHz}}$. L04 p16
2. FSK: $B = (1 + d) \times S + 2\Delta f$. L04 p15
3. $100 \text{ kHz} = (1 + 1) \times S + 50 \text{ kHz} \Rightarrow S = (100 \text{ kHz} - 50 \text{ kHz}) / 2 = 25 \text{ kbaud}$.
4. Binary FSK has $r = 1$, so $N = S = \mathbf{25 \text{ kbps}}$.
5. Bit 1 $\rightarrow f_c + \Delta f = \mathbf{275 \text{ kHz}}$; bit 0 $\rightarrow f_c - \Delta f = \mathbf{225 \text{ kHz}}$. L04 p14

Carrier $f_c = 250 \text{ kHz}$ Bit rate $N = 25 \text{ kbps}$ f_1 (bit 1) = **275 kHz** f_2 (bit 0) = **225 kHz**

Slide's answer: $f_c = 250 \text{ kHz}$, $N = 25 \text{ kbps}$

✏ WORKED EXAMPLE

MFSK: 3 bits at a time L04 p19

We need to send data **3 bits at a time** at a bit rate of **3 Mbps**. The carrier frequency is **10 MHz**. Using MFSK with the minimum carrier spacing ($2\Delta f = S$) and $d = 0$, calculate the number of levels (different frequencies), the baud rate, the bandwidth, and the lowest and highest carrier frequencies.

1. $L = 2^r = 2^3 = \mathbf{8}$ frequencies. L04 p19
2. $S = N / r = 3 \text{ Mbps} / 3 = \mathbf{1 \text{ Mbaud}}$, so the carriers are $2\Delta f = 1 \text{ MHz}$ apart.
3. $B = (1 + d) \times S + (L - 1) \times 2\Delta f = 1 \text{ MHz} + 7 \times 1 \text{ MHz} = \mathbf{8 \text{ MHz}}$ ($= L \times S$). L04 p18
4. The 8 carriers sit symmetrically around f_c : from **6.5 MHz** to **13.5 MHz**. L04 p20

L (frequencies) = **8** Baud rate $S = \mathbf{1 \text{ Mbaud}}$ Bandwidth $B = \mathbf{8 \text{ MHz}}$ Lowest carrier = **6.5 MHz**

Highest carrier = **13.5 MHz**

Slide's answer: $L = 8$, $S = 1 \text{ Mbaud}$, $B = 8 \text{ MHz}$ (carriers 6.5 ... 13.5 MHz, slide 20)

Phase shift keying: BPSK and QPSK L04 pp21–26

PSK changes the **phase** of the carrier; amplitude and frequency stay constant. Its bandwidth is the same as ASK: $B = (1 + d) \times S$. PSK is **much more robust than ASK**, because noise mostly changes amplitude. L04 p21

- **BPSK**: two elements 180° apart — bit 1 → 0°, bit 0 → 180°. L04 p22 Implementation: multiply a **polar NRZ** signal by the carrier (BASK uses *unipolar* NRZ). L04 p23
- **QPSK**: 2 bits per element ($L = 4$). The stream is split into pairs: the **first bit** drives the **in-phase (I)** carrier and the **second bit** the **quadrature (Q)** carrier, which is shifted 90°. The two BPSK signals are added. L04 p24 L04 p25

Dibit	I (1st bit)	Q (2nd bit)	Phase
11	+	+	45°
01	–	+	135°
00	–	–	–135° (225°)
10	+	–	–45° (315°)

WORKED EXAMPLE

QPSK bandwidth L04 p26

Find the baud rate and the bandwidth for a signal transmitting at **12 Mbps** using **QPSK**. The value of $d = 0$.

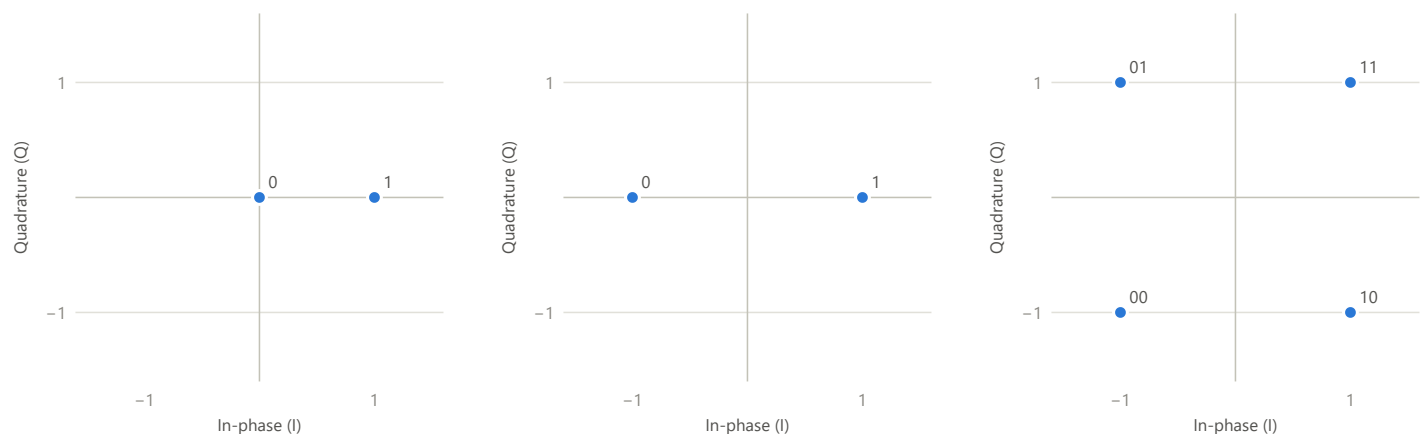
1. QPSK carries $r = \log_2 4 = 2$ bits per signal element. L04 p24
2. $S = N \times 1/r = 12 \text{ Mbps} / 2 = \mathbf{6 \text{ Mbaud}}$. L04 p26
3. $B = (1 + d) \times S = 1 \times 6 \text{ MHz} = \mathbf{6 \text{ MHz}}$. L04 p26

Baud rate $S = \mathbf{6 \text{ Mbaud}}$ Bandwidth $B = \mathbf{6 \text{ MHz}}$

Slide's answer: $S = 6 \text{ Mbaud}$, $B = 6 \text{ MHz}$

Constellation diagrams L04 pp27–29

A constellation diagram defines the amplitude and phase of each signal element when two carriers are used, one in quadrature with the other: the **x-axis is the in-phase (I) carrier** and the **y-axis the quadrature (Q) carrier**. L04 p27 For any point, the **length** of the line from the origin is the **amplitude** and its **angle** is the **phase**. L04 p28



ASK (OOK): 0 at the origin, 1 on +I.

BPSK: 0 at -I, 1 at +I (180° apart).

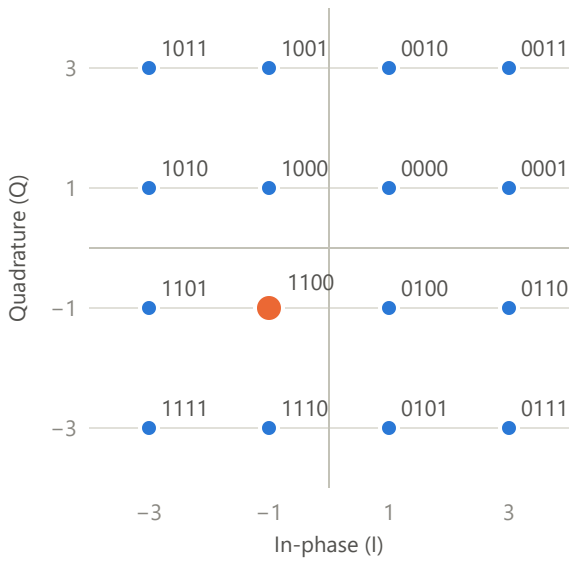
QPSK: four points 90° apart.

Going counter-clockwise from 45°, the QPSK labels run **11 → 01 → 00 → 10**. L04 p29 **Beyond slides** Neighbours differ in one bit (Gray coding), so a small phase error flips at most one bit.

Quadrature amplitude modulation (QAM) L04 pp30–31

QAM is a combination of **ASK and PSK**: both amplitude and phase change, using the I and Q carriers. Slide 30 shows several 4-QAM layouts — four unipolar points, four points at (±1, ±1) (the same as QPSK), or four points in one quadrant — and a 16-QAM layout. L04 p30

The labelled **16-QAM** on slide 31 puts I and Q at ±25% and ±75% of full scale. The **first two bits pick the quadrant** (00 upper right, 10 upper left, 11 lower left, 01 lower right) and the last two pick the point inside it. 16-QAM carries $r = \log_2 16 = 4$ bits per element. L04 p31



16-QAM from slide 31 (1 unit = 25%). Highlighted: 1100 at I = Q = -25%, phase 225°.

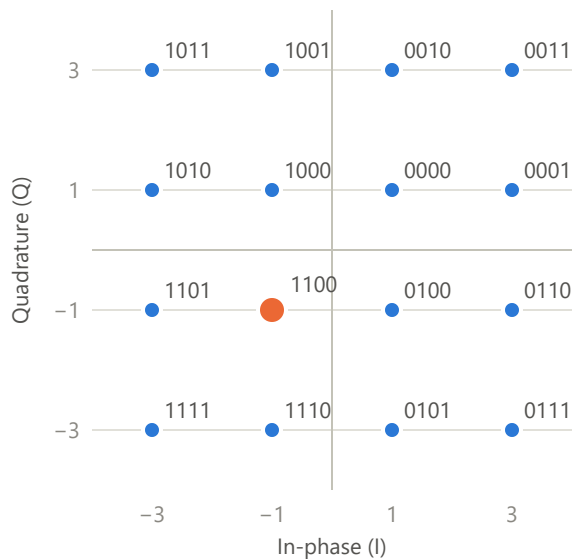
Q ↓ / I →	-75%	-25%	+25%	+75%
+75%	1011	1001	0010	0011
+25%	1010	1000	0000	0001
-25%	1101	1100	0100	0110
-75%	1111	1110	0101	0111

WORKED EXAMPLE

Reading the 16-QAM constellation L04 p31

In the slides' 16-QAM constellation (1 unit = 25% of full scale), what is the **phase** of the signal element that carries **1100**?

1. Locate 1100: it sits at $I = -1$, $Q = -1$. L04 p31
2. The angle from the +I axis is the phase: $\text{atan2}(Q, I) = -135^\circ$ (same as 225°).
3. Its distance from the origin is the amplitude: $\sqrt{I^2 + Q^2} = 1.414$ units. L04 p28



Phase = -135°

Slide's answer: 225° (slide inset)

Formulas

Baud (signal) rate L04 p5

$$S = N \times \frac{1}{r}$$

S in baud; N = bit rate (bps); r = data bits per signal element

Analog transmission of digital data: $S \leq N$.

Bits per element $\leftrightarrow$ levels L04 p7

$$r = \log_2 L \Leftrightarrow L = 2^r$$

L = number of distinct signal elements (levels, phases, frequencies)

ASK / PSK bandwidth L04 p9; L04 p21

$$B = (1 + d) \times S$$

$0 \leq d \leq 1$ (modulation and filtering); $d = 0$ gives the minimum, $B = S$

Binary FSK bandwidth L04 p14; L04 p15

$$B = (1 + d) \times S + 2\Delta f$$

$$2\Delta f = f_1 - f_2; \text{ bit } 1 \rightarrow f_c + \Delta f, \text{ bit } 0 \rightarrow f_c - \Delta f$$

Multilevel FSK bandwidth L04 p18; L04 p19

$$B = (1 + d) \times S + (L - 1) \times 2\Delta f$$

with $d = 0$ and the minimum spacing $2\Delta f = S$: $B = L \times S$

⚠ Slide 18 prints " $(L - 1)/2\Delta f$ " — it is a product.

Carrier in an available band L04 p12; L04 p13

$$f_c = (f_{\text{low}} + f_{\text{high}}) / 2$$

full duplex: split the band in two halves, one carrier in the middle of each

Reading a constellation point L04 p28

$$A = \sqrt{I^2 + Q^2}, \quad \theta = \text{angle from the +I axis}$$

distance from the origin = amplitude; angle = phase

⚠ COMMON TRAPS

- **Plugging the bit rate N into $B = (1 + d)S$** → Bandwidth uses the **baud rate**. Convert first: $S = N / r$. L04 p9
- **MFSK: dividing by $2\Delta f$ because slide 18 prints “ $(L - 1)/2\Delta f$ ”** → It is $(L - 1) \times 2\Delta f$; with the minimum spacing, $B = L \times S$. L04 p18
- **FSK: forgetting the $2\Delta f$ term** → Subtract $2\Delta f$ from B before dividing by $(1 + d)$. L04 p15
- **Full duplex: giving one direction the whole band** → Split the band in two; each direction gets half and its own carrier. L04 p13
- **QPSK: sending the first bit on Q** → Slide convention: first bit → I (x-axis), second bit → Q (y-axis). 01 is at 135° . L04 p25
- **Calling ASK as robust as PSK** → Noise mostly changes amplitude, so ASK suffers more; PSK is much more robust than ASK. L04 p21
- **Taking 25% as the amplitude of 16-QAM's 1100** → 25% is each axis component; the distance from the origin is $\sqrt{2} \times 25\% \approx 35\%$. The 225° phase is right. L04 p31

💡 MNEMONICS & MEMORY HOOKS

- ASK, FSK, PSK = **A**mplitude, **F**requency, **P**hase — the three characteristics of a sine wave from L02.
- “Bits ride on bauds”: **$N = r \times S$** . More bits per element → more data on the same bandwidth (ASK, PSK, QAM).
- **$L = 2^r$** : $2 \rightarrow 1$ bit, $4 \rightarrow 2$, $8 \rightarrow 3$, $16 \rightarrow 4$, $256 \rightarrow 8$.
- QPSK counter-clockwise from 45° : **11, 01, 00, 10** — each step changes one bit.
- FSK: “one is high” — bit 1 uses $f_c + \Delta f$.

Blank-page recall

Close the notes. Answer each prompt from memory on paper, then check against the sections above.

1. Write the relationship between N, S, r and L, then redo slide 7 ($N = 8000$ bps, $S = 1000$ baud).
2. Write the bandwidth formulas for ASK, binary FSK, MFSK and PSK. Which ones depend on Δf ?
3. Sketch the OOK, BPSK and QPSK constellations with the slide labels.
4. Explain why PSK is more robust than ASK, and why QPSK doubles BPSK's bit rate at the same baud rate.
5. Solve without notes: band 200–300 kHz, ASK, $d = 1$, full duplex. Carriers? Bit rate per direction?

⚠ SLIDE ERRORS & CONTRADICTIONS

On the exam, the answer that follows the slide's intent is the safe one — but know the correct version.

Slide	Slide says	Correct / intended
L04 p18	$B = (1+d) \times S + (L-1)/2\Delta f = L \times S$	$B = (1+d) \times S + (L-1) \times 2\Delta f$; it equals $L \times S$ only when $d = 0$ and $2\Delta f = S$.
L04 p14	The figure writes Δf as “Df” (and the extracted text drops Δ entirely)	Read “2Df” as $2\Delta f$, the spacing between f_1 and f_2 .
L04 p14	Text: bit 1 $\rightarrow f_1 = f_c + \Delta f$, but the spectrum draws f_1 below f_2	Follow the text: bit 1 uses the higher frequency $f_c + \Delta f$; bit 0 uses $f_c - \Delta f$.
L04 p31	Inset: point 1100 has amplitude 25%	25% is the I and the Q component; the magnitude is $\sqrt{2} \times 25\% \approx 35\%$. The 225° phase is correct.
L04 p15	A stray “5.15” label	A leftover textbook slide number — ignore it.

Glossary

Carrier signal	The analog sine wave (frequency f_c) whose amplitude, frequency or phase is changed to carry the data. L04 p2
Bit rate (N)	Bits sent per second (bps). L04 p5
Baud rate (S)	Signal elements sent per second (baud); also the signal or modulation rate. $S = N/r$. L04 p5
ASK	Amplitude shift keying — the carrier's amplitude encodes the data. $B = (1 + d)S$. L04 p8; L04 p9
OOK	On-off keying: binary ASK where a 1 sends the carrier and a 0 sends nothing. L04 p8; L04 p10
d (modulation factor)	A value between 0 and 1 that depends on modulation and filtering; it widens the bandwidth from S to $(1 + d)S$. L04 p9
FSK	Frequency shift keying — the carrier's frequency encodes the data. $B = (1 + d)S + 2\Delta f$. L04 p14; L04 p15
Coherent FSK	FSK that switches frequency only at the same phase of the signal, so the waveform stays continuous. L04 p17
Non-coherent FSK	FSK that switches frequency without adhering to the current phase. L04 p17
MFSK	Multilevel FSK: $L = 2^r$ frequencies, each carrying r bits. L04 p18
VCO	Voltage-controlled oscillator — produces a frequency that follows its input voltage; used to implement FSK. L04 p18
PSK	Phase shift keying — the carrier's phase encodes the data. $B = (1 + d)S$; more noise-robust than ASK. L04 p21
BPSK	Binary PSK: two elements 180° apart (bit 1 $\rightarrow 0^\circ$, bit 0 $\rightarrow 180^\circ$); polar NRZ $\times$ carrier. L04 p22; L04 p23
QPSK	Quadrature PSK: 2 bits per element on I and Q carriers 90° apart; $L = 4$. L04 p24
In-phase carrier (I)	The reference carrier; the x-axis of a constellation diagram. L04 p27

Quadrature carrier (Q)

The carrier shifted 90° from the in-phase carrier; the y-axis of a constellation diagram.

L04 p24; L04 p27

Constellation diagram

A plot of the signal elements: distance from the origin = amplitude, angle = phase.

L04 p27; L04 p28

QAM

Quadrature amplitude modulation — a combination of ASK and PSK (amplitude and phase both change). L04 p30

END OF DOCUMENT: reviewer · build r1